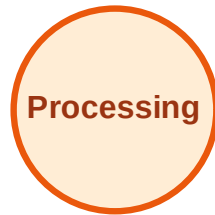


BFS Traversal

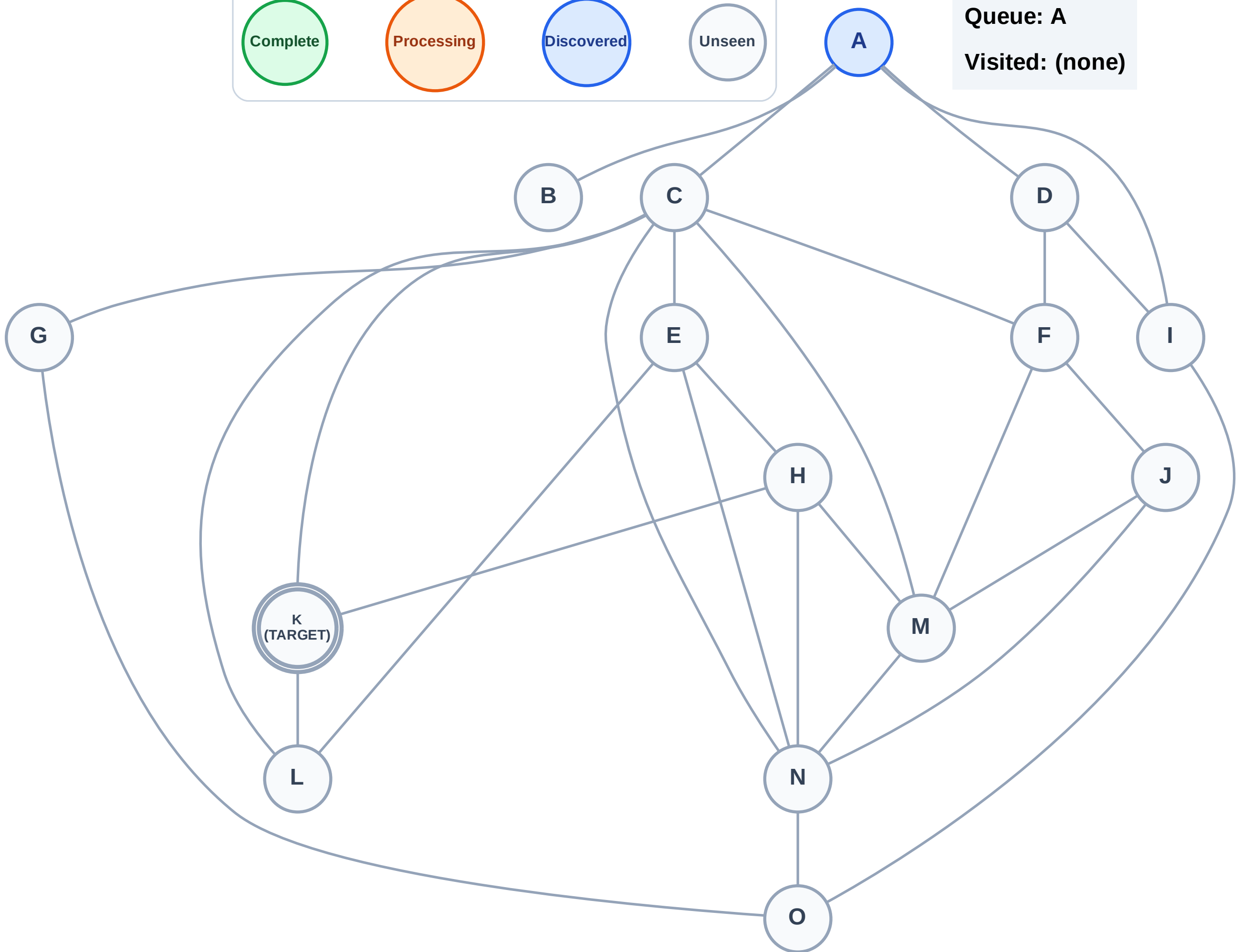
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

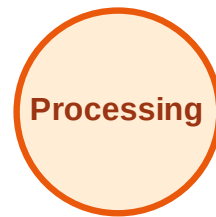
Visited: (none)



BFS Traversal

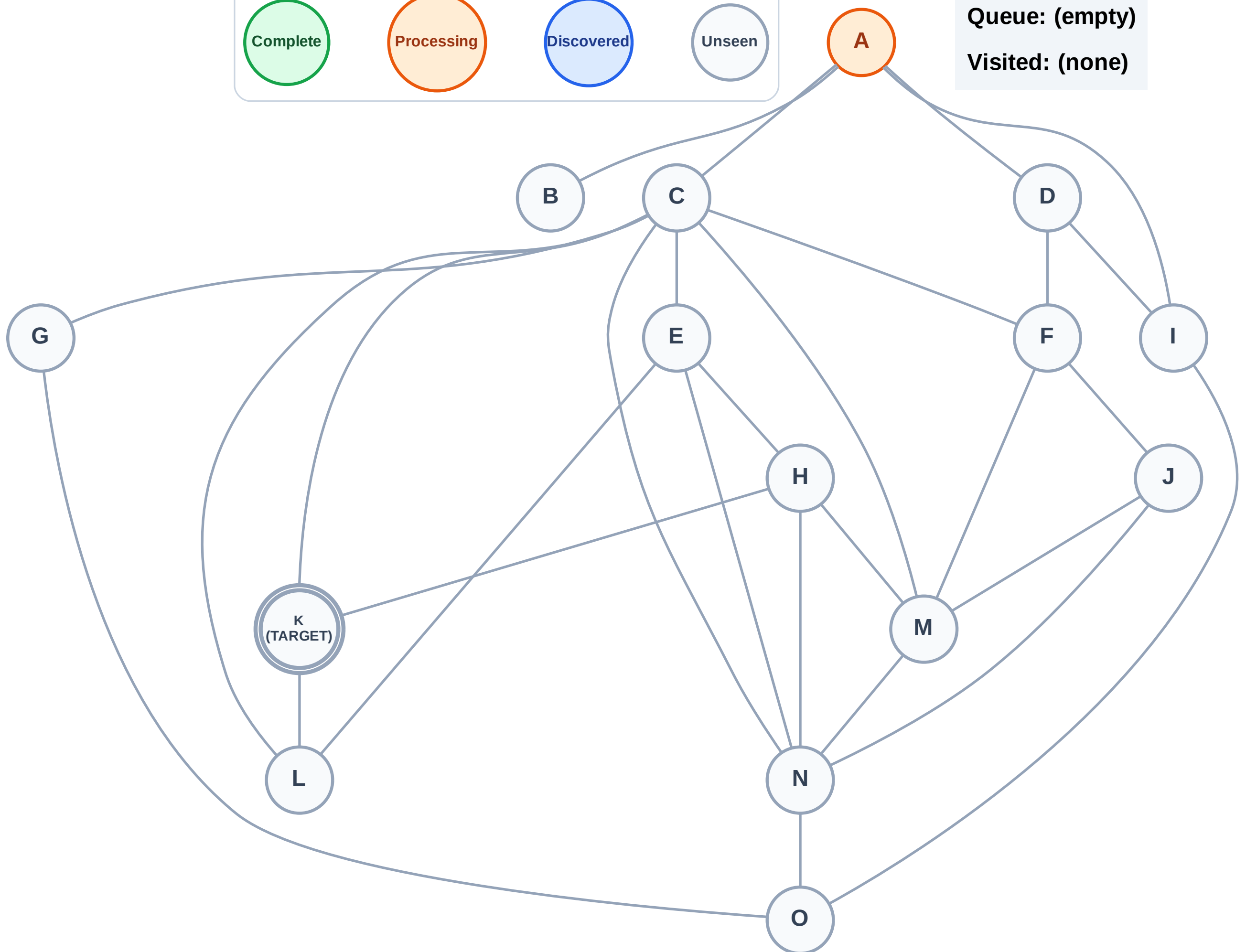
Step 2: Process node A

Legend



Queue: (empty)

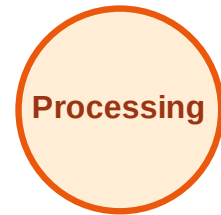
Visited: (none)



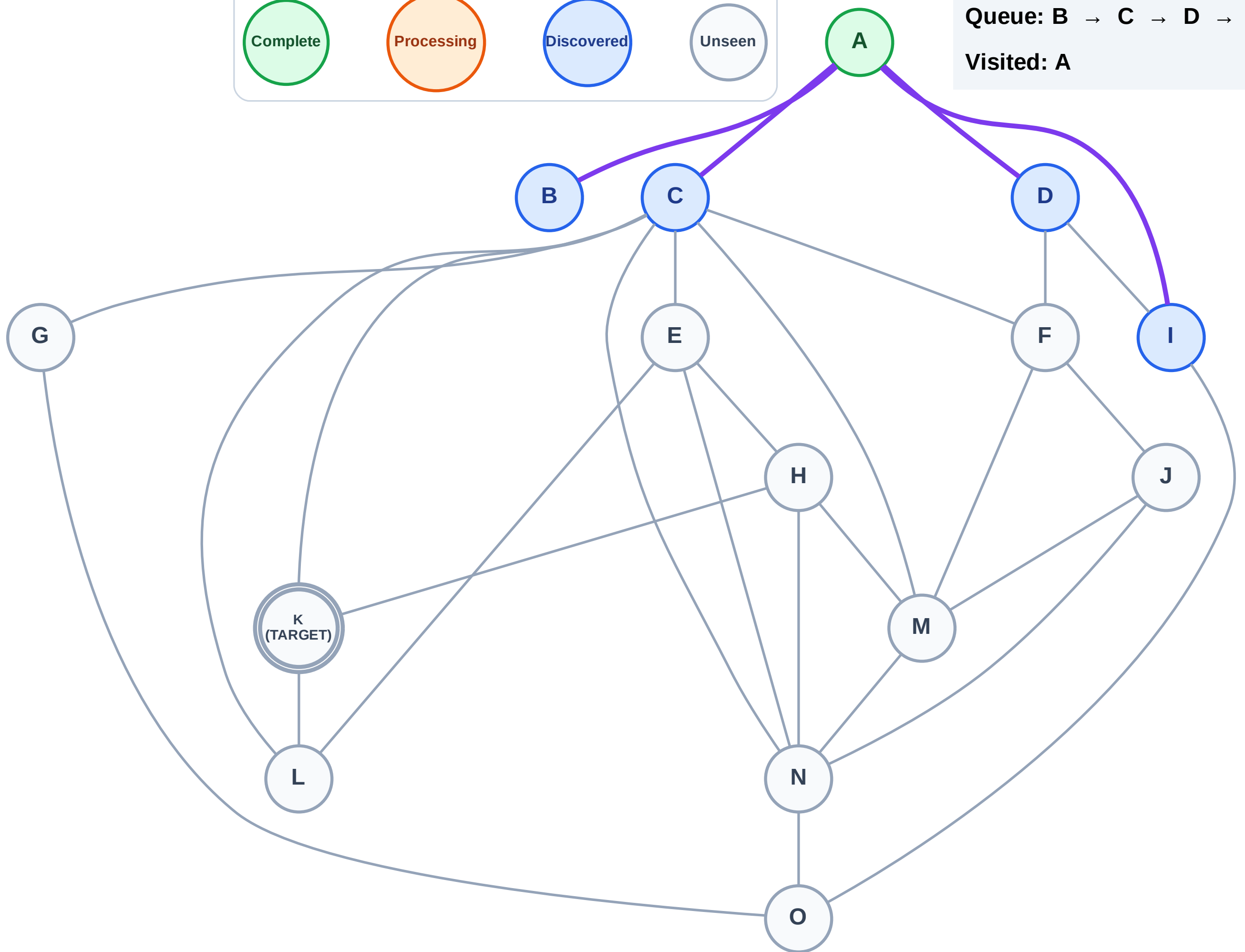
BFS Traversal

Step 3: Discover B, C, D, I from A

Legend



Queue: B → C → D → I
Visited: A



Step 4: Process node B

Step 4: Process node B

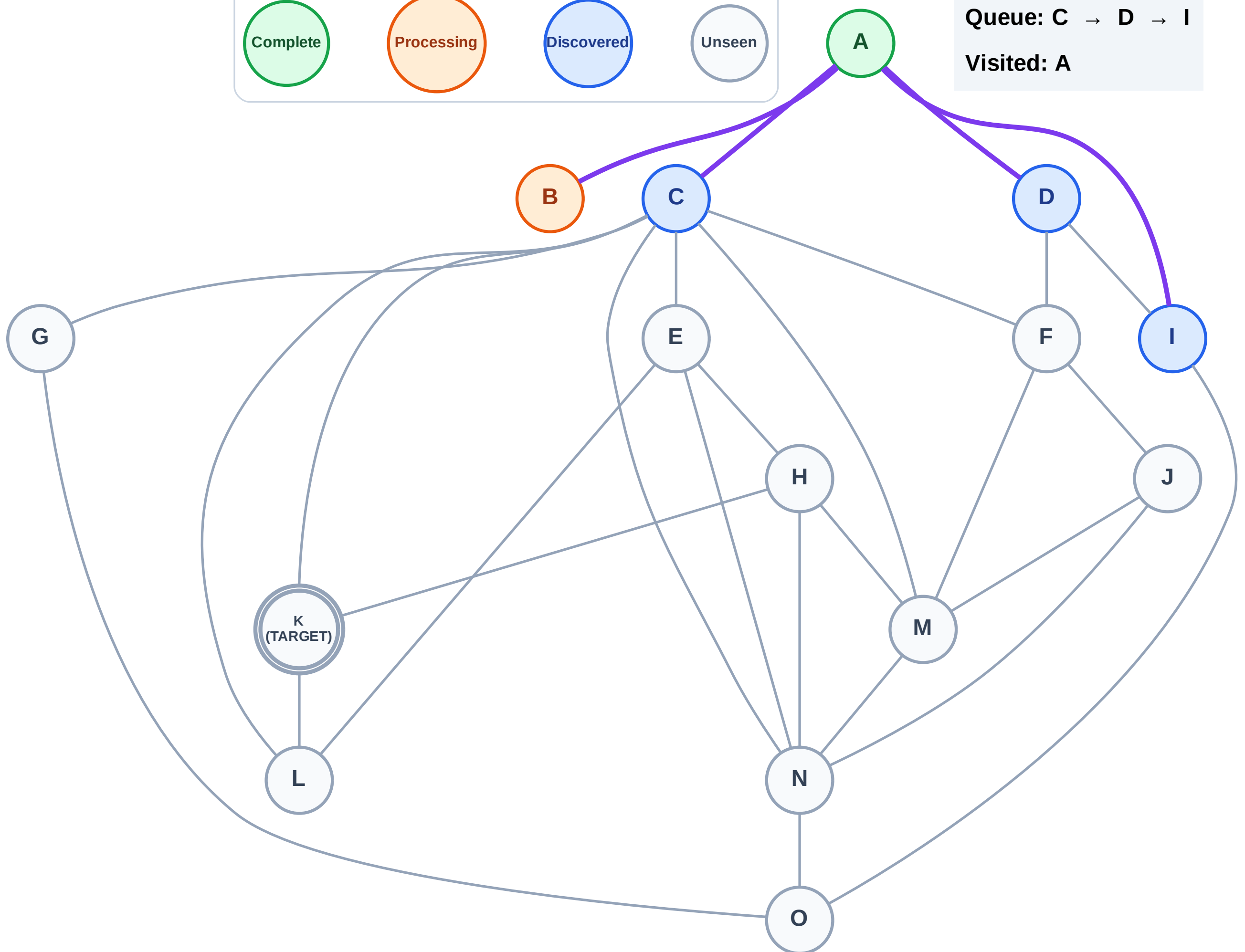
The diagram illustrates the four stages of the data science process, each represented by a colored circle:

- Complete** (Green circle)
- Processing** (Orange circle)
- Discovered** (Blue circle)
- Unseen** (Light blue circle)

Processing

Unseen

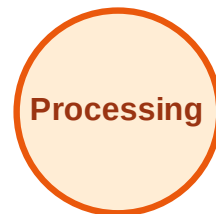
Visited: A



BFS Traversal

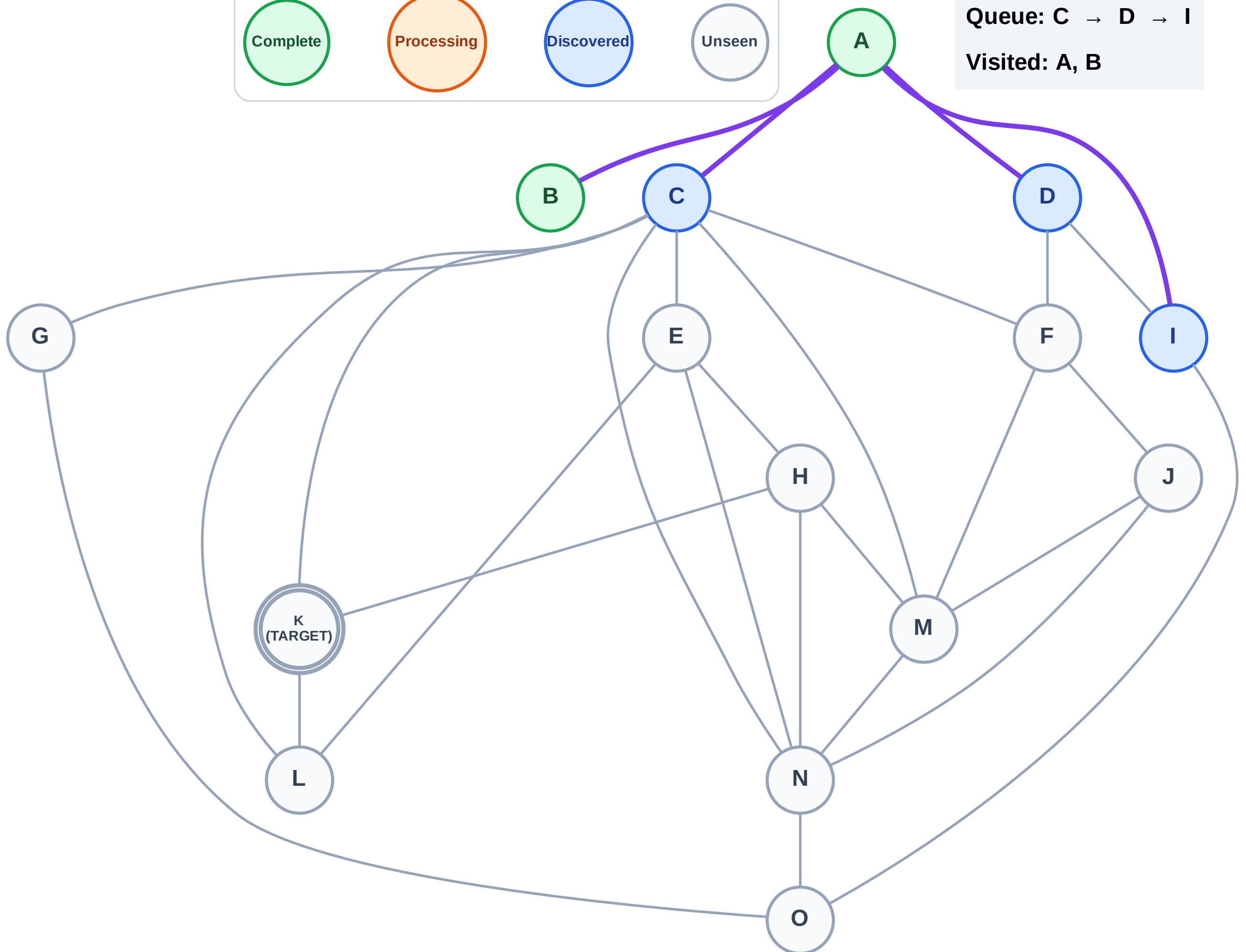
Step 5: No new neighbors from B

Legend



Queue: C → D → I

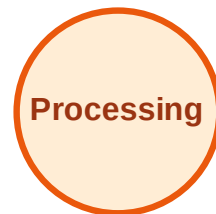
Visited: A, B



BFS Traversal

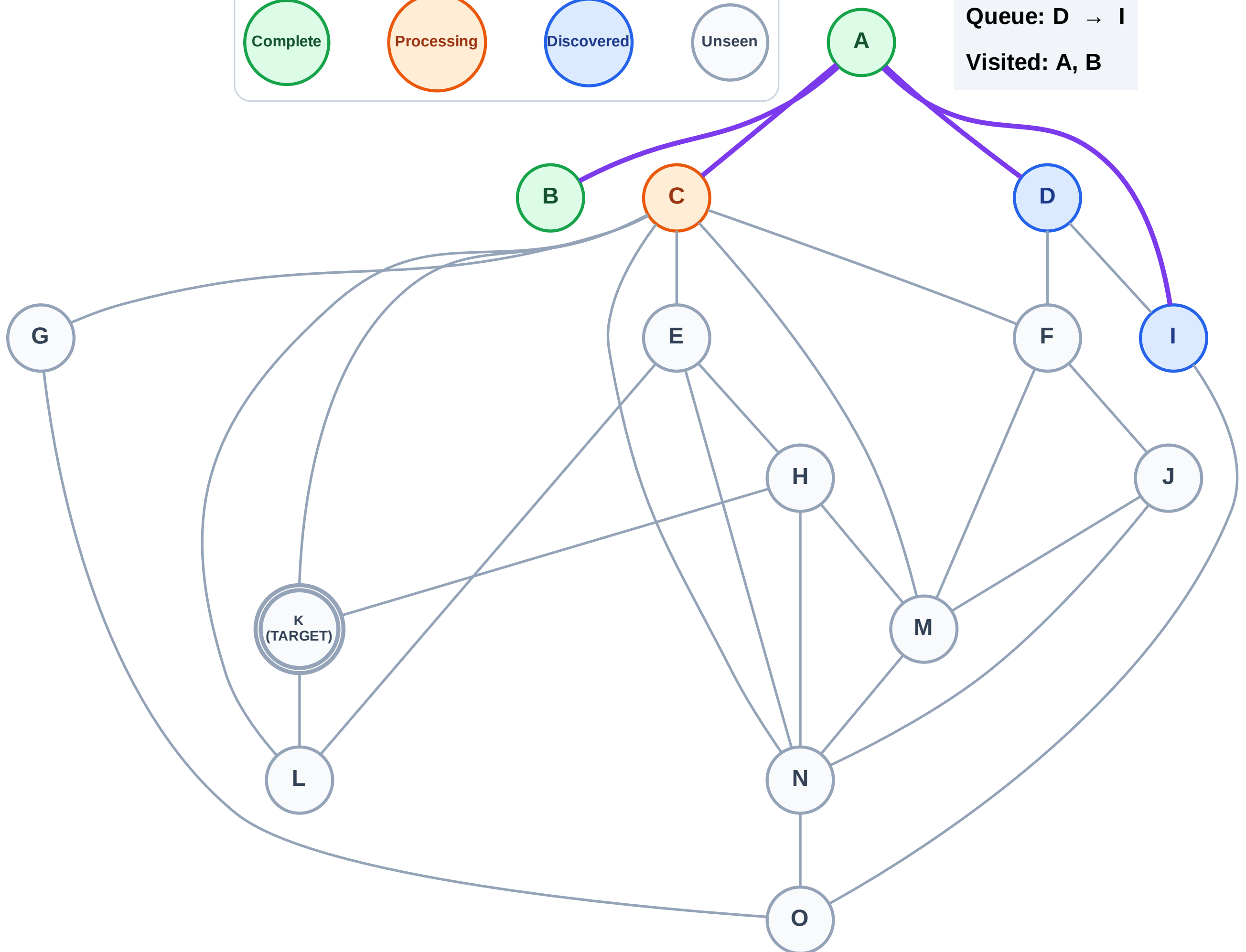
Step 6: Process node C

Legend



Queue: D → I

Visited: A, B



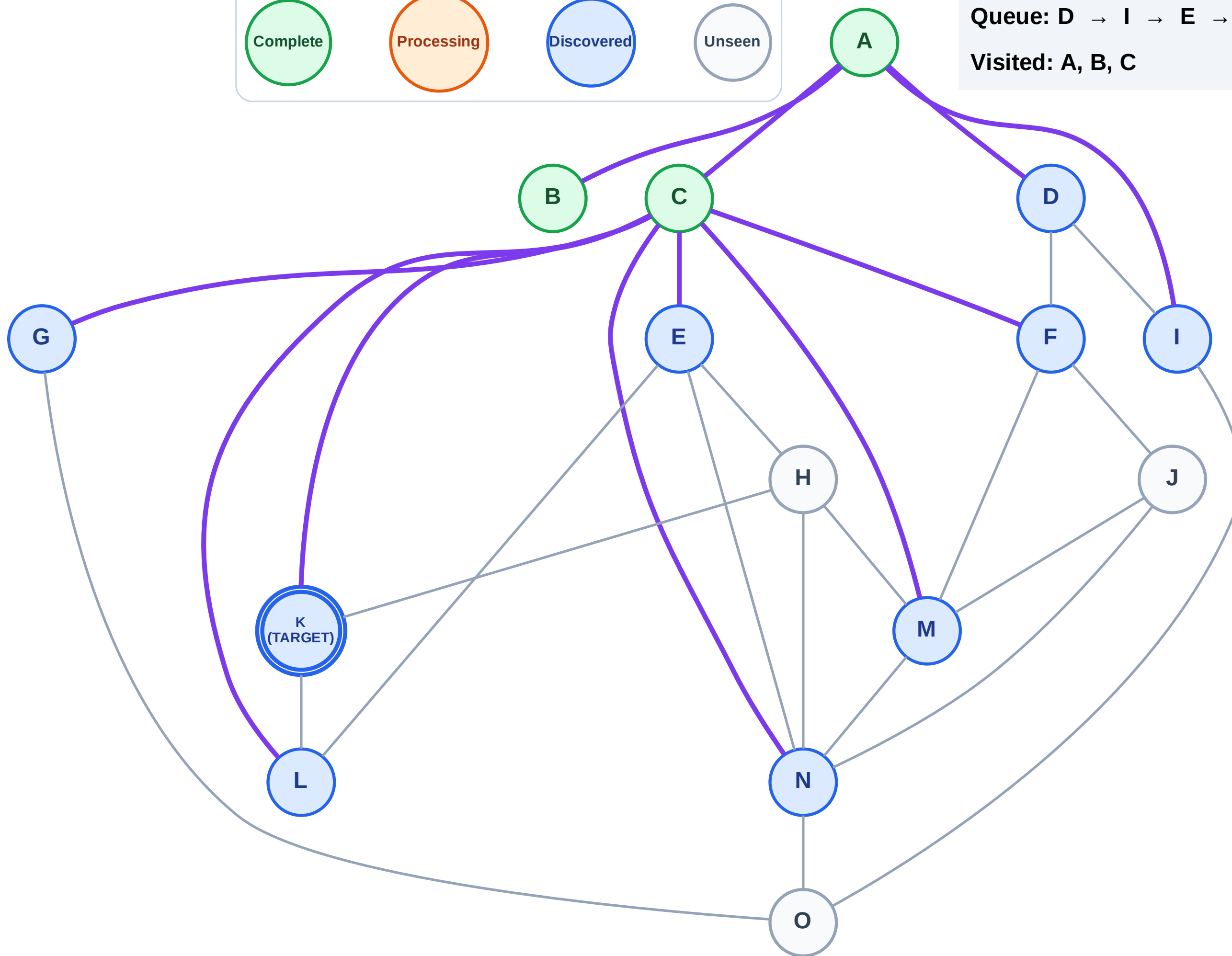
BFS Traversal

Step 7: Discover E, F, G, K, L, M, N from C

Legend



Queue: D → I → E → F → G → K → L → M → N
Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

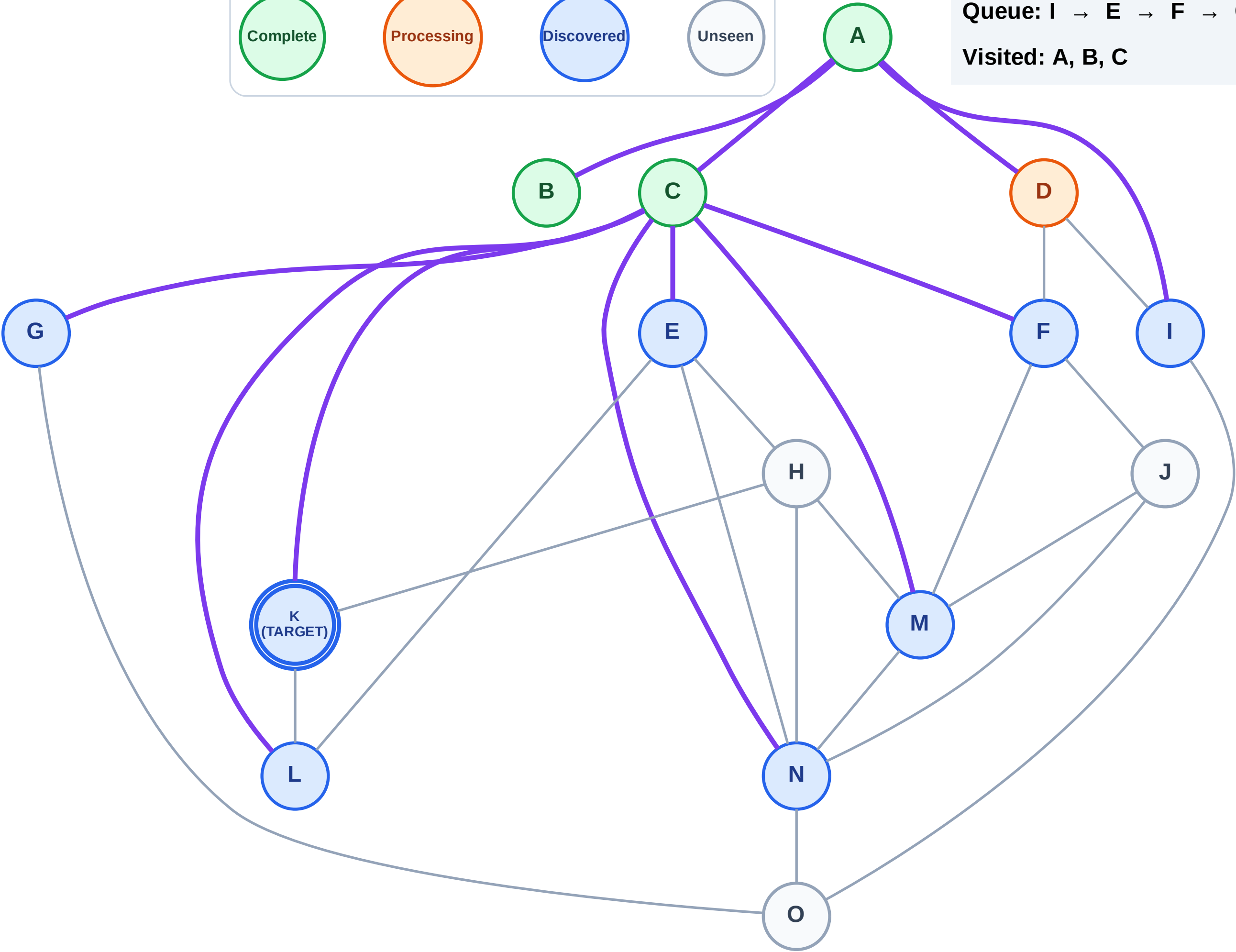
Processing

Discovered

Unseen

Queue: I → E → F → G → K → L → M → N

Visited: A, B, C



BFS Traversal

Step 9: No new neighbors from D

Legend

Complete

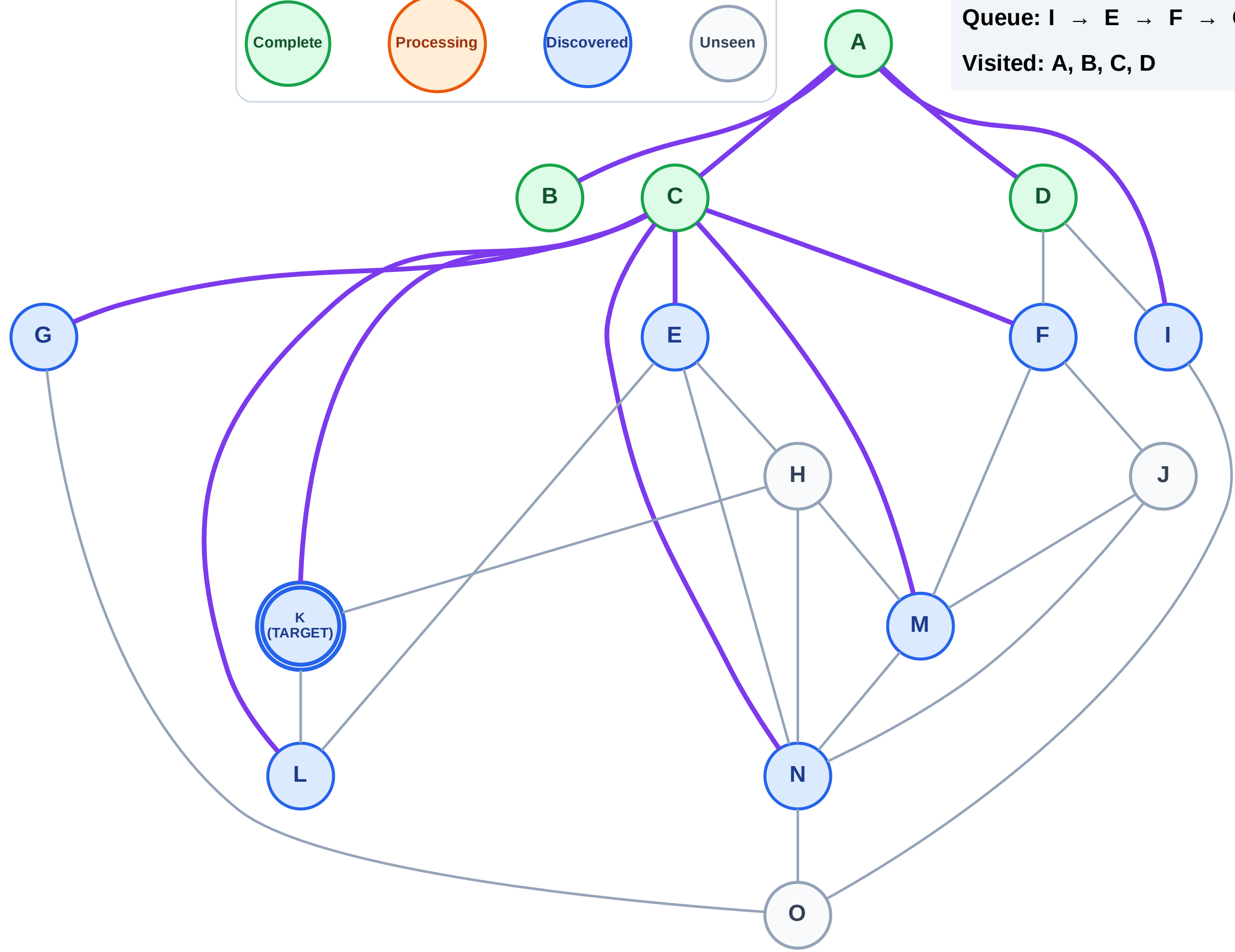
Processing

Discovered

Unseen

Queue: I → E → F → G → K → L → M → N

Visited: A, B, C, D



BFS Traversal

Step 10: Process node I

Legend

Complete

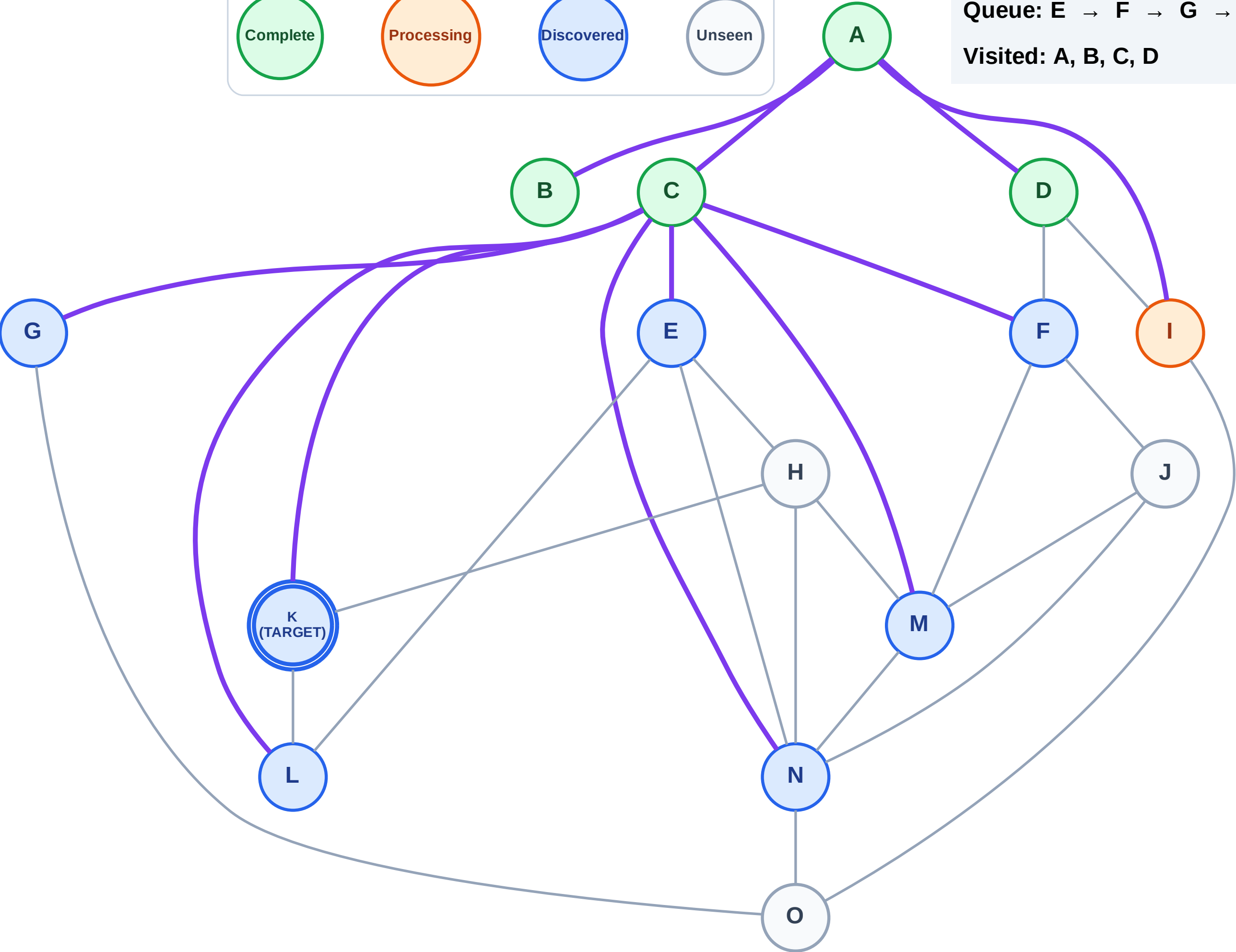
Processing

Discovered

Unseen

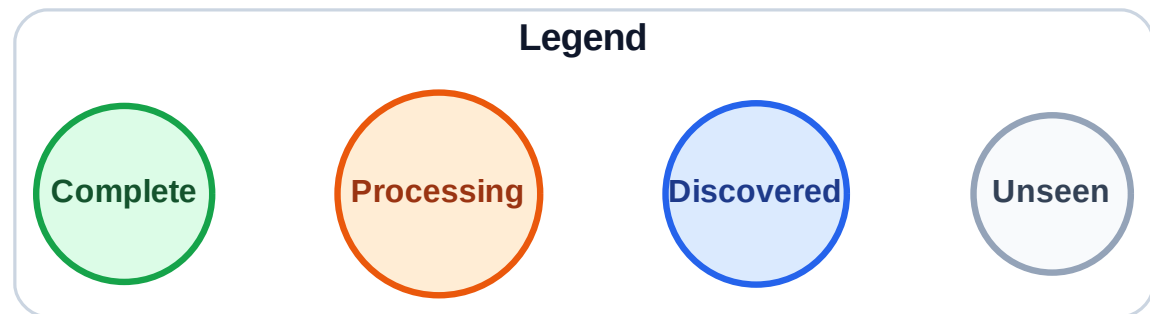
Queue: E → F → G → K → L → M → N

Visited: A, B, C, D



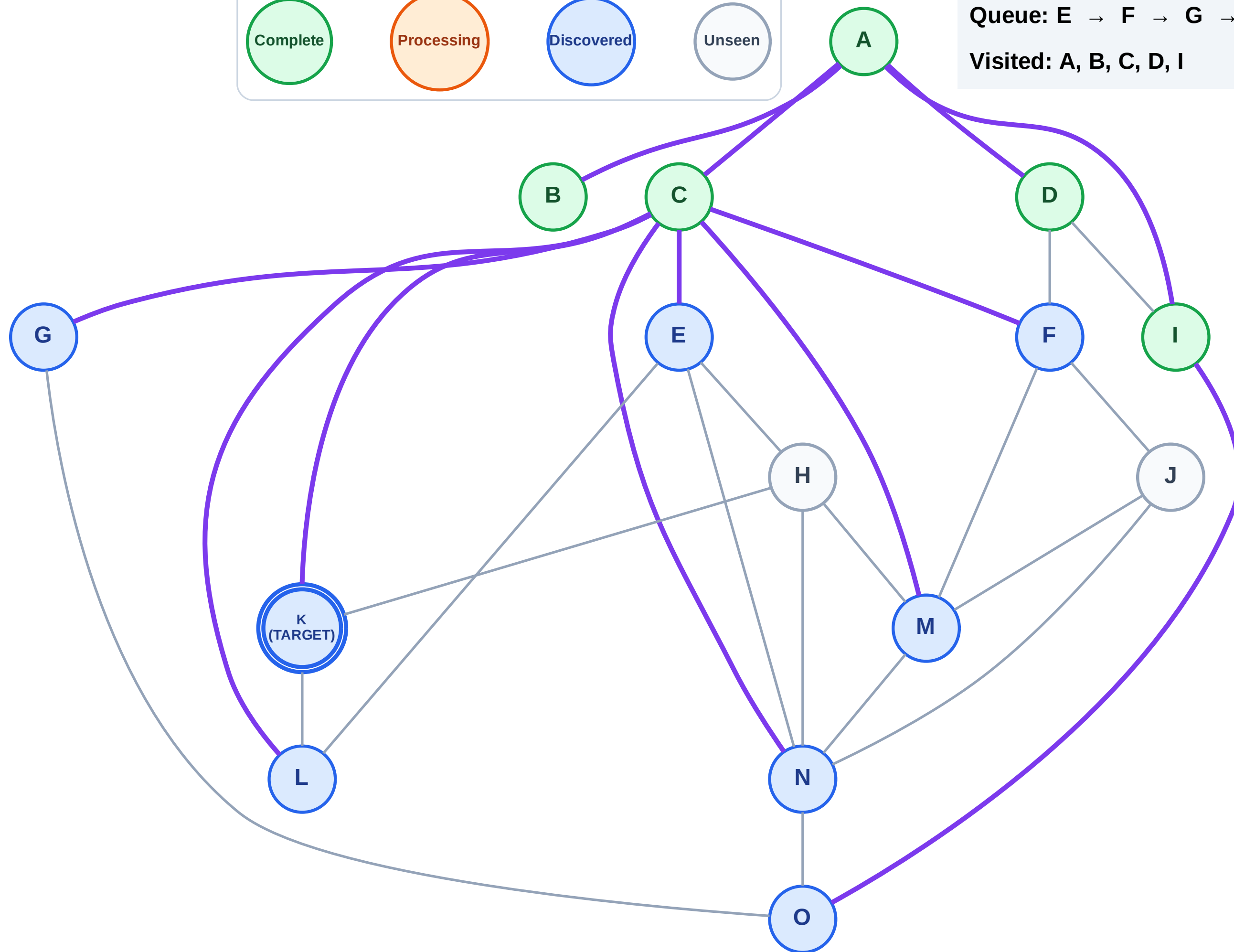
BFS Traversal

Step 11: Discover O from I



Queue: E → F → G → K → L → M → N → O

Visited: A, B, C, D, I



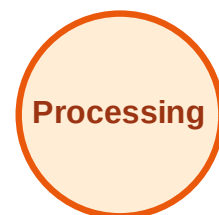
BFS Traversal

Step 12: Process node E

Legend



Complete



Processing



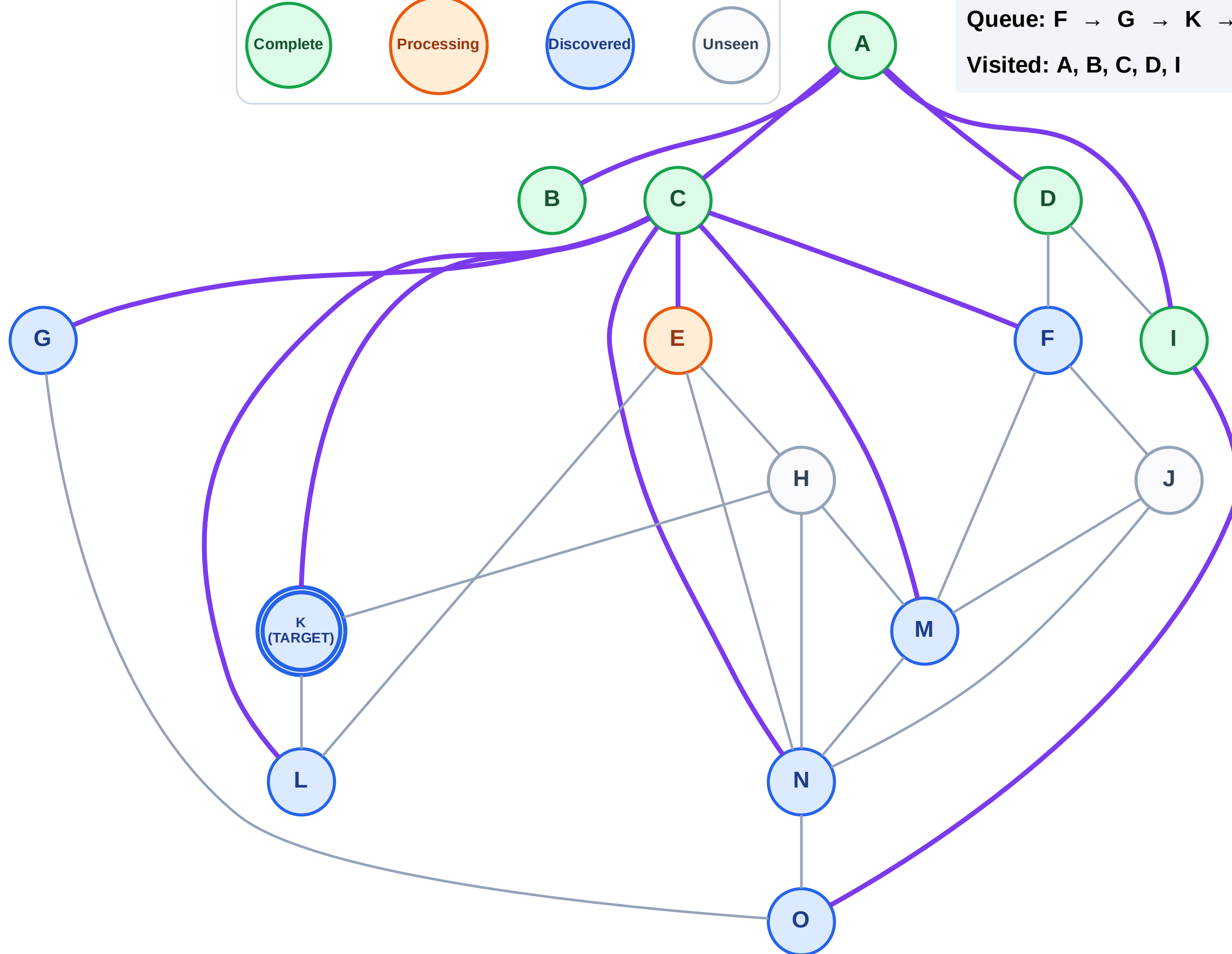
Discovered



Unseen

Queue: F → G → K → L → M → N → O

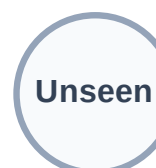
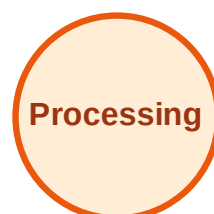
Visited: A, B, C, D, I



BFS Traversal

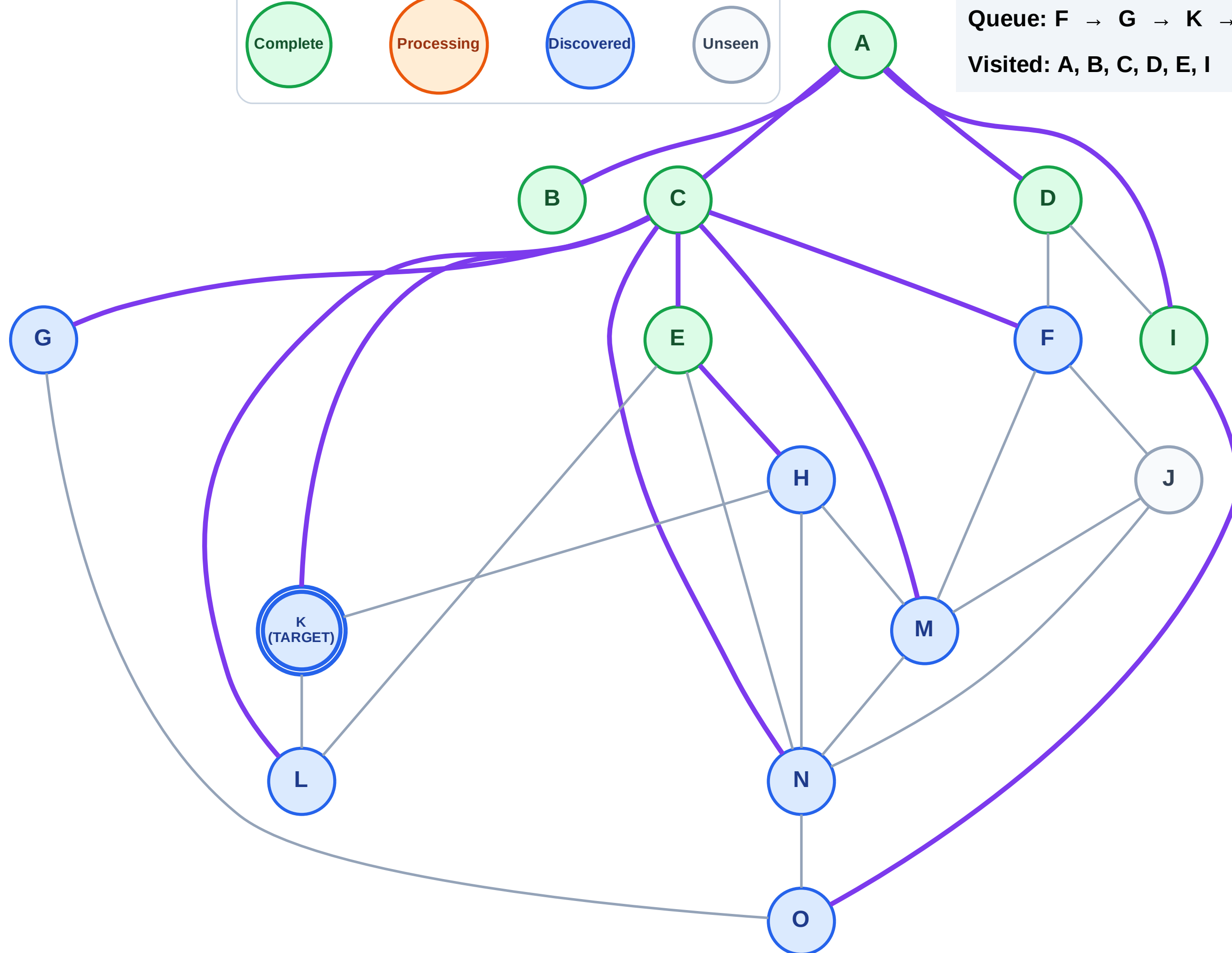
Step 13: Discover H from E

Legend



Queue: F → G → K → L → M → N → O → H

Visited: A, B, C, D, E, I



BFS Traversal

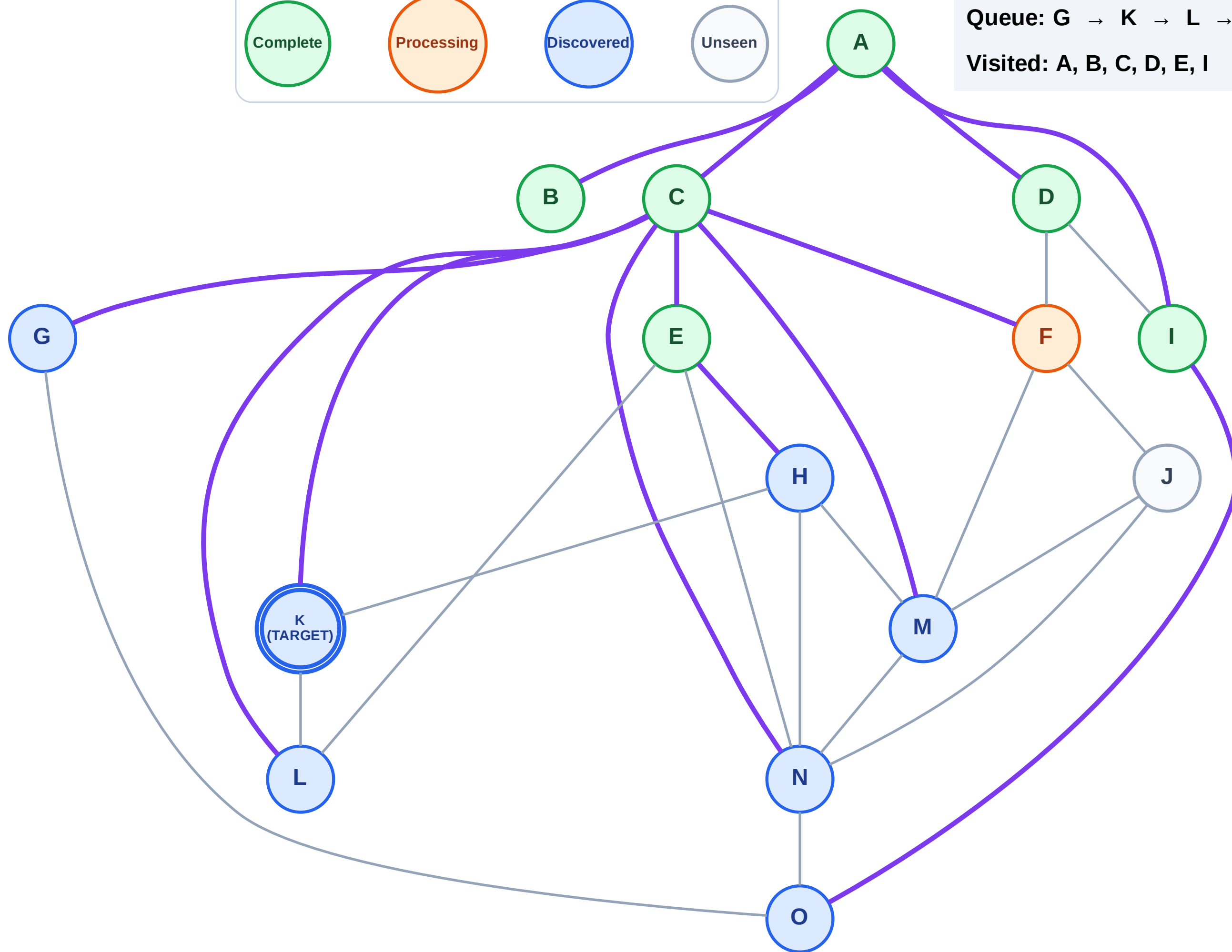
Step 14: Process node F

Legend



Queue: G → K → L → M → N → O → H

Visited: A, B, C, D, E, I



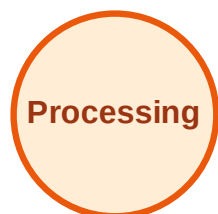
BFS Traversal

Step 15: Discover J from F

Legend



Complete



Processing



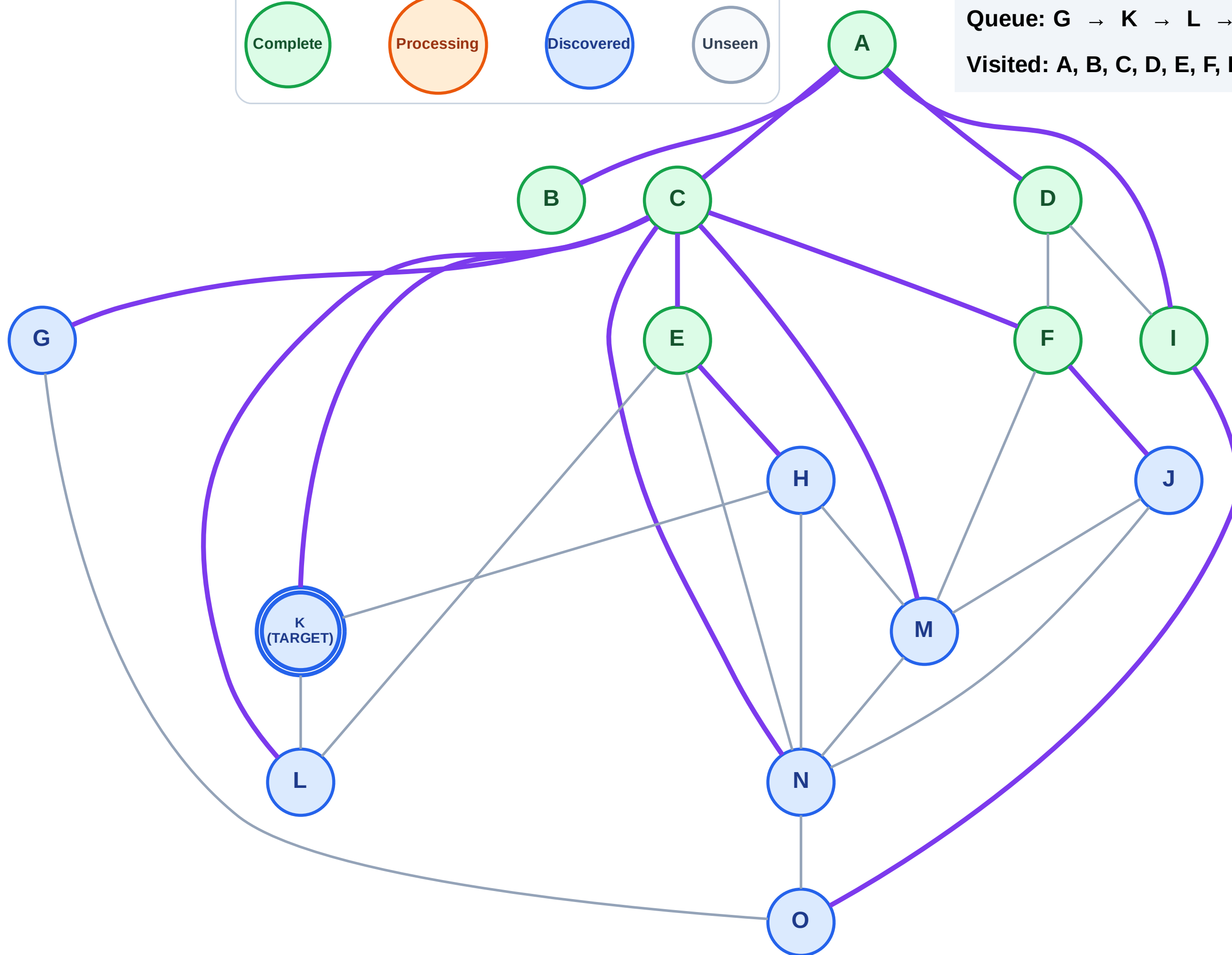
Discovered



Unseen

Queue: G → K → L → M → N → O → H → J

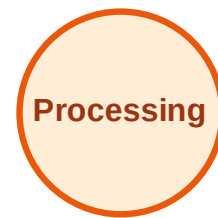
Visited: A, B, C, D, E, F, I



BFS Traversal

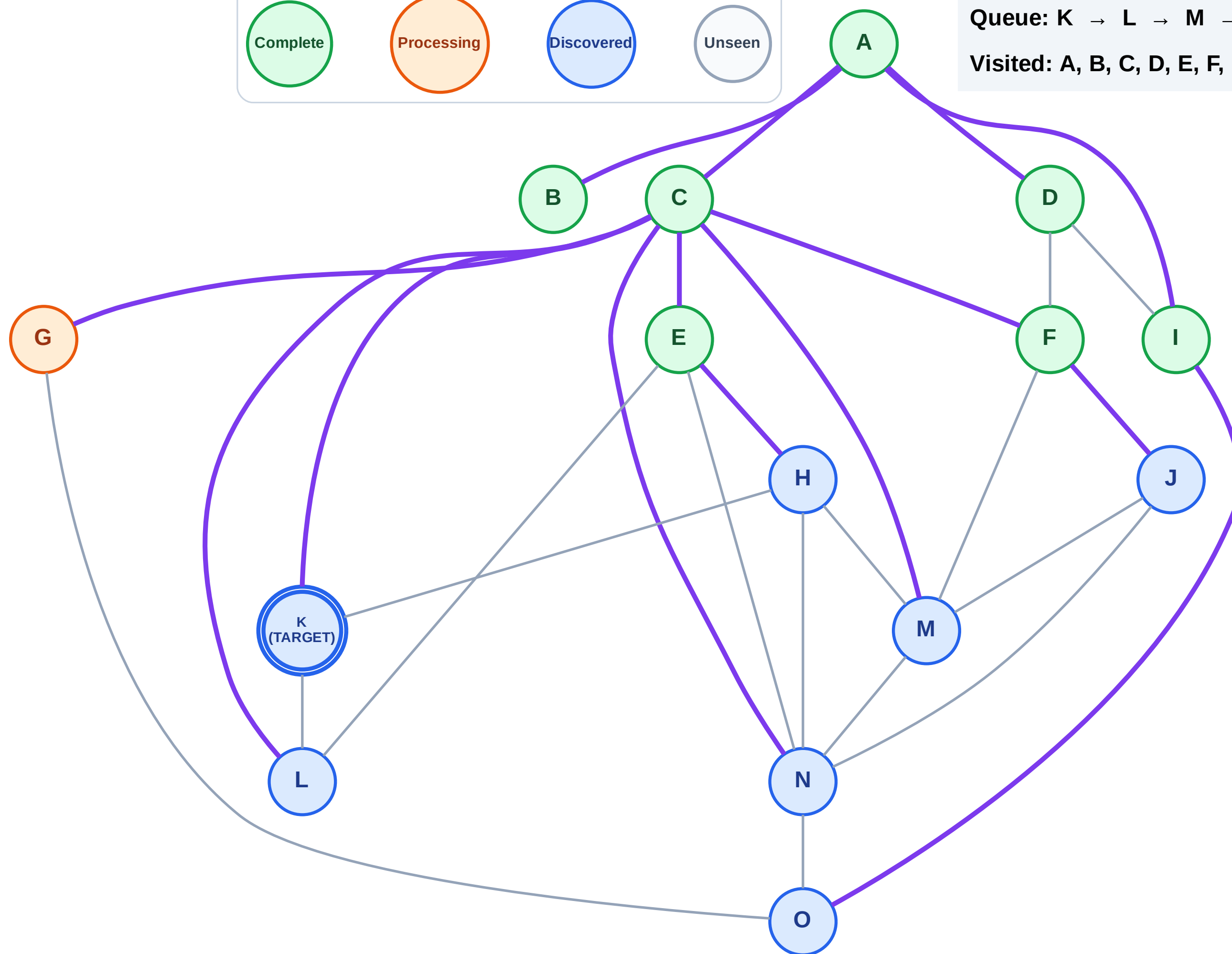
Step 16: Process node G

Legend



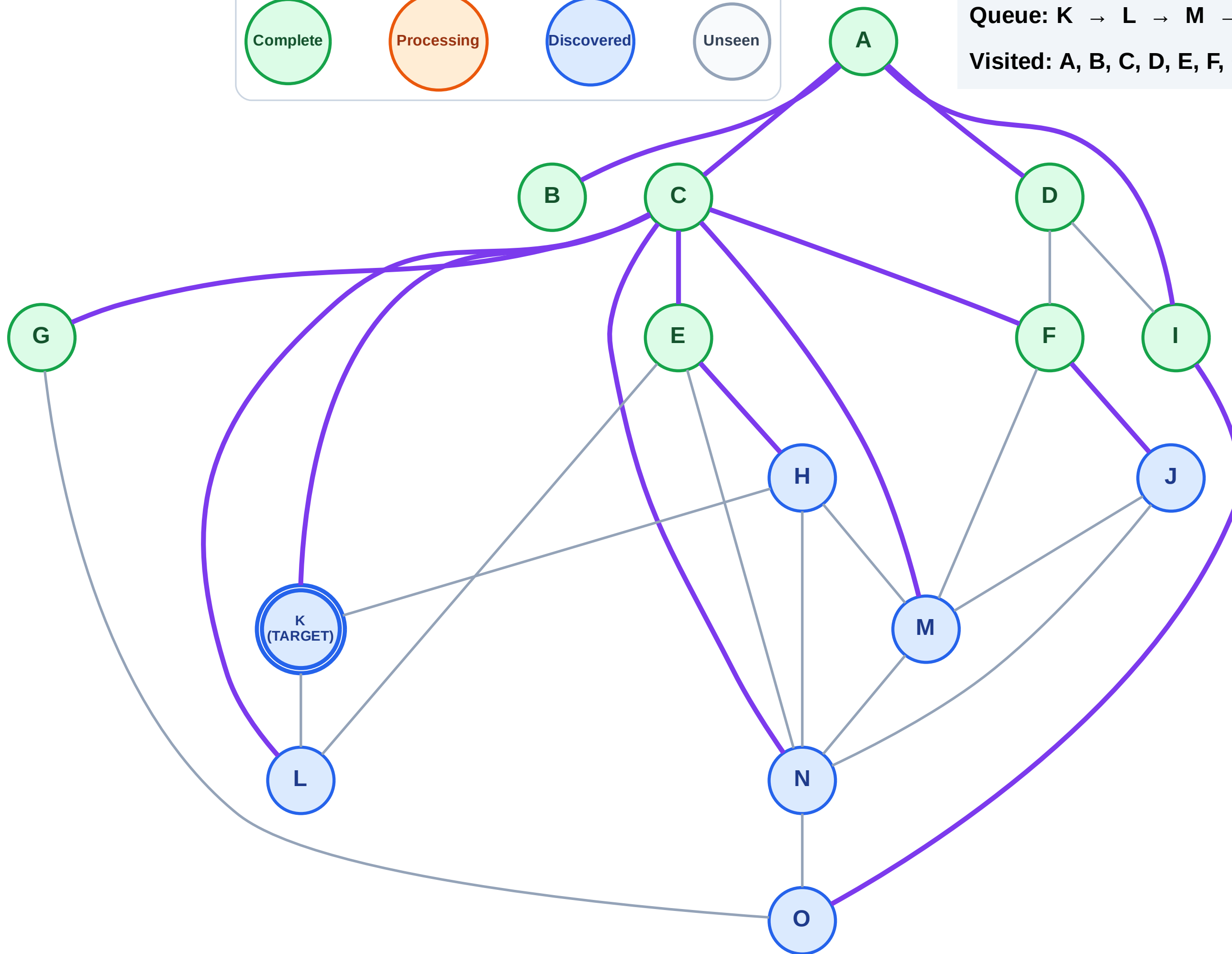
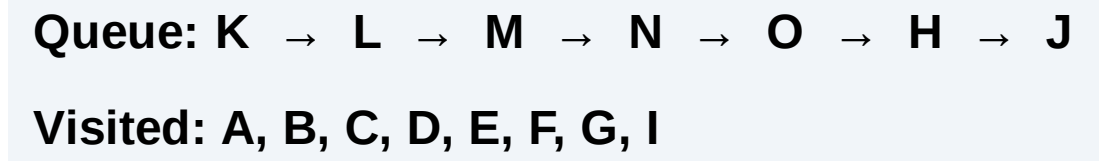
Queue: K → L → M → N → O → H → J

Visited: A, B, C, D, E, F, I



Step 17: No new neighbors from G

Legend



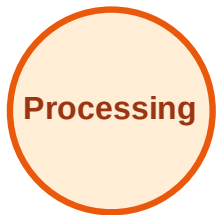
BFS Traversal

Step 18: Process node K

Legend



Processing



Discovered

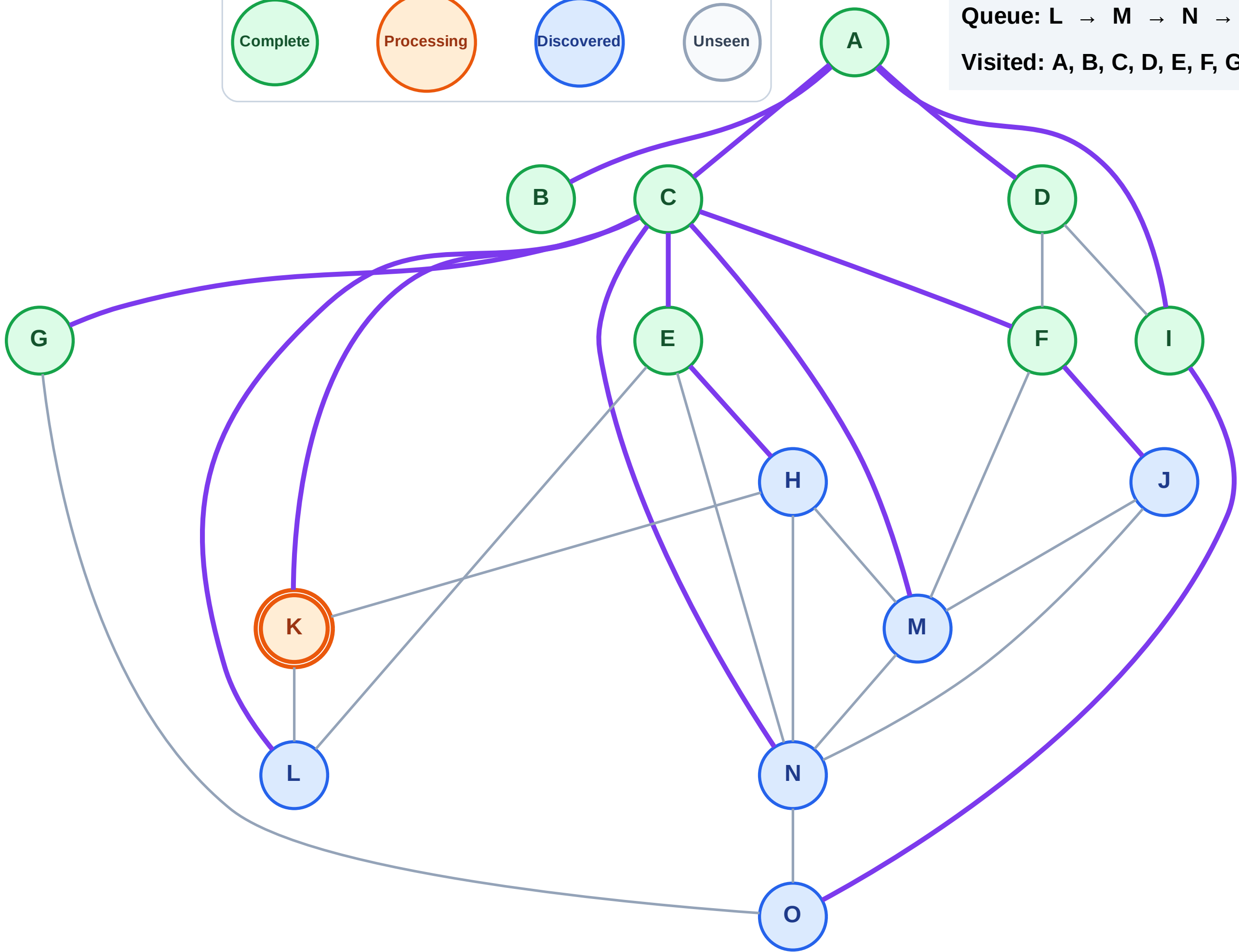


Unseen



Queue: L → M → N → O → H → J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 19: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend

Complete

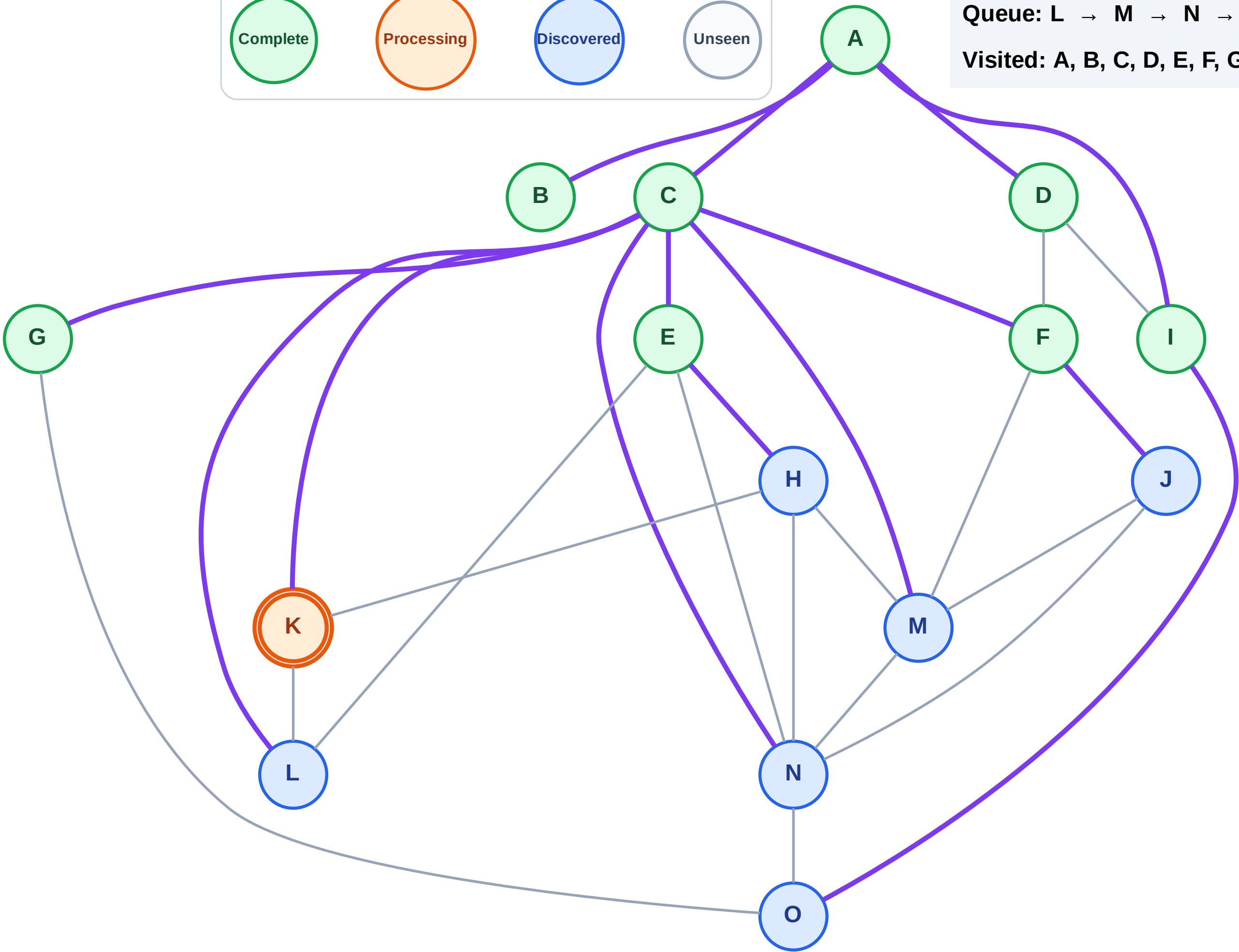
Processing

Discovered

Unseen

Queue: L → M → N → O → H → J

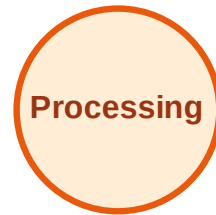
Visited: A, B, C, D, E, F, G, I, K



DFS Traversal

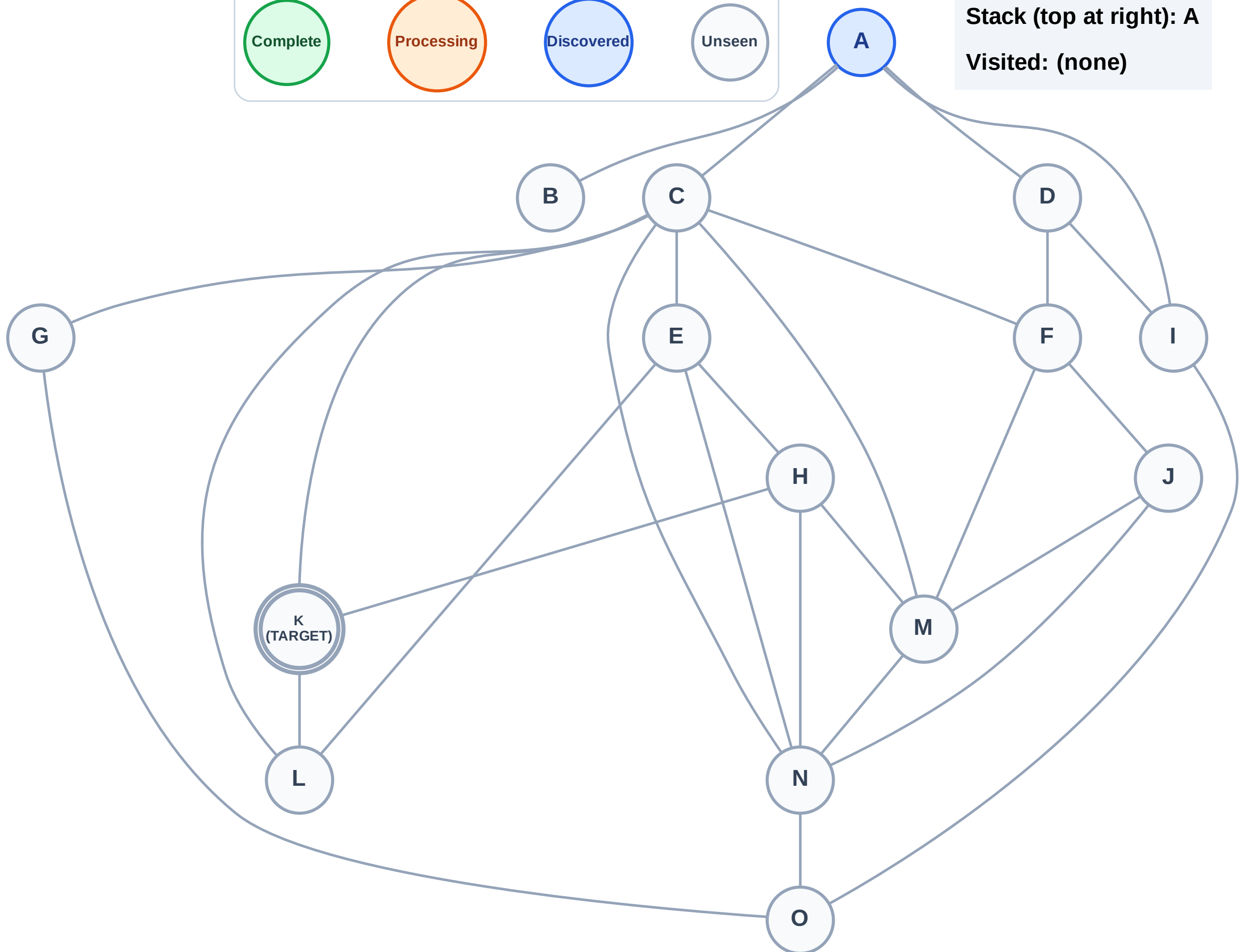
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

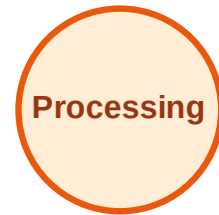
Visited: (none)



DFS Traversal

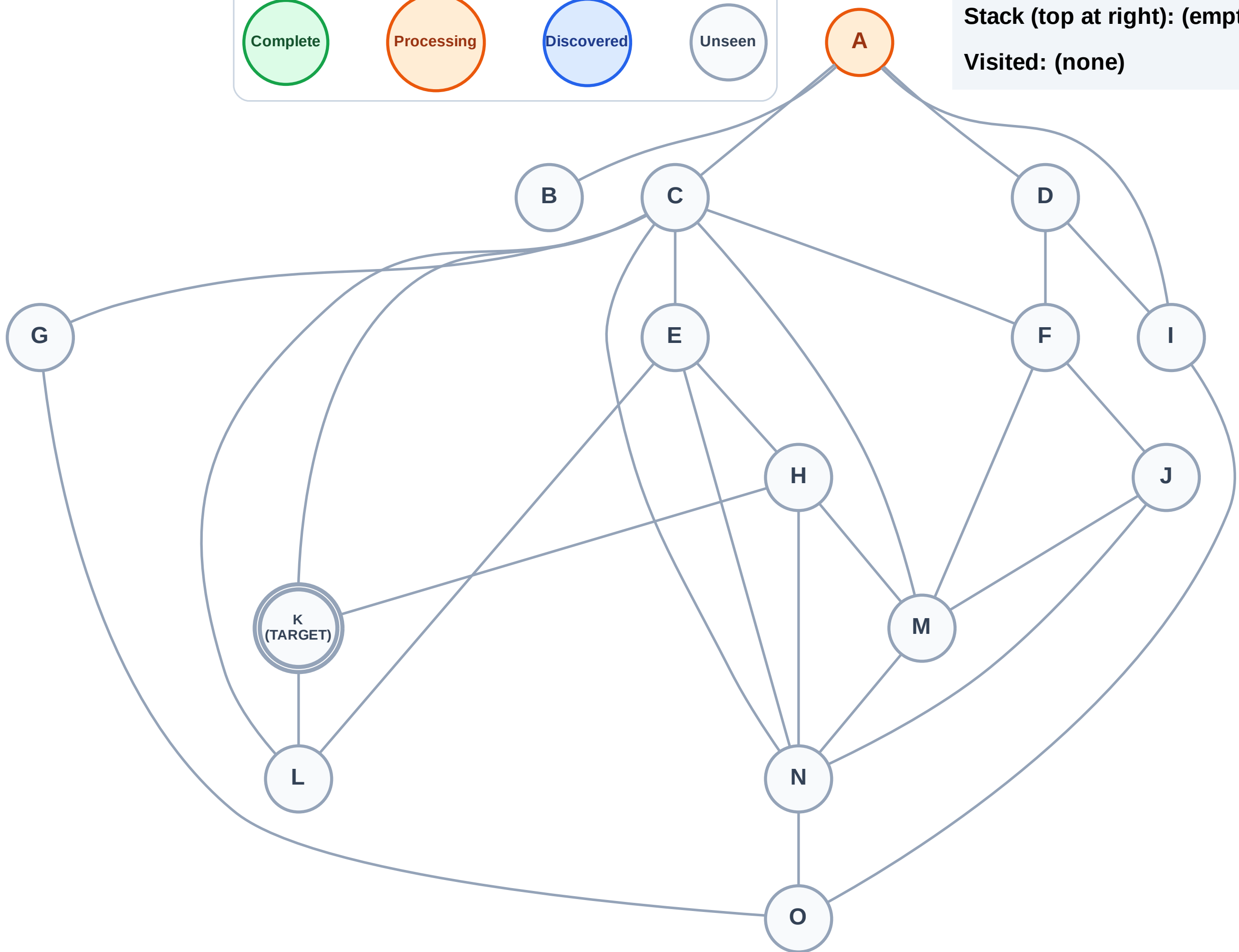
Step 2: Process node A

Legend



Stack (top at right): (empty)

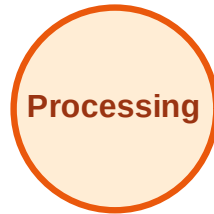
Visited: (none)



DFS Traversal

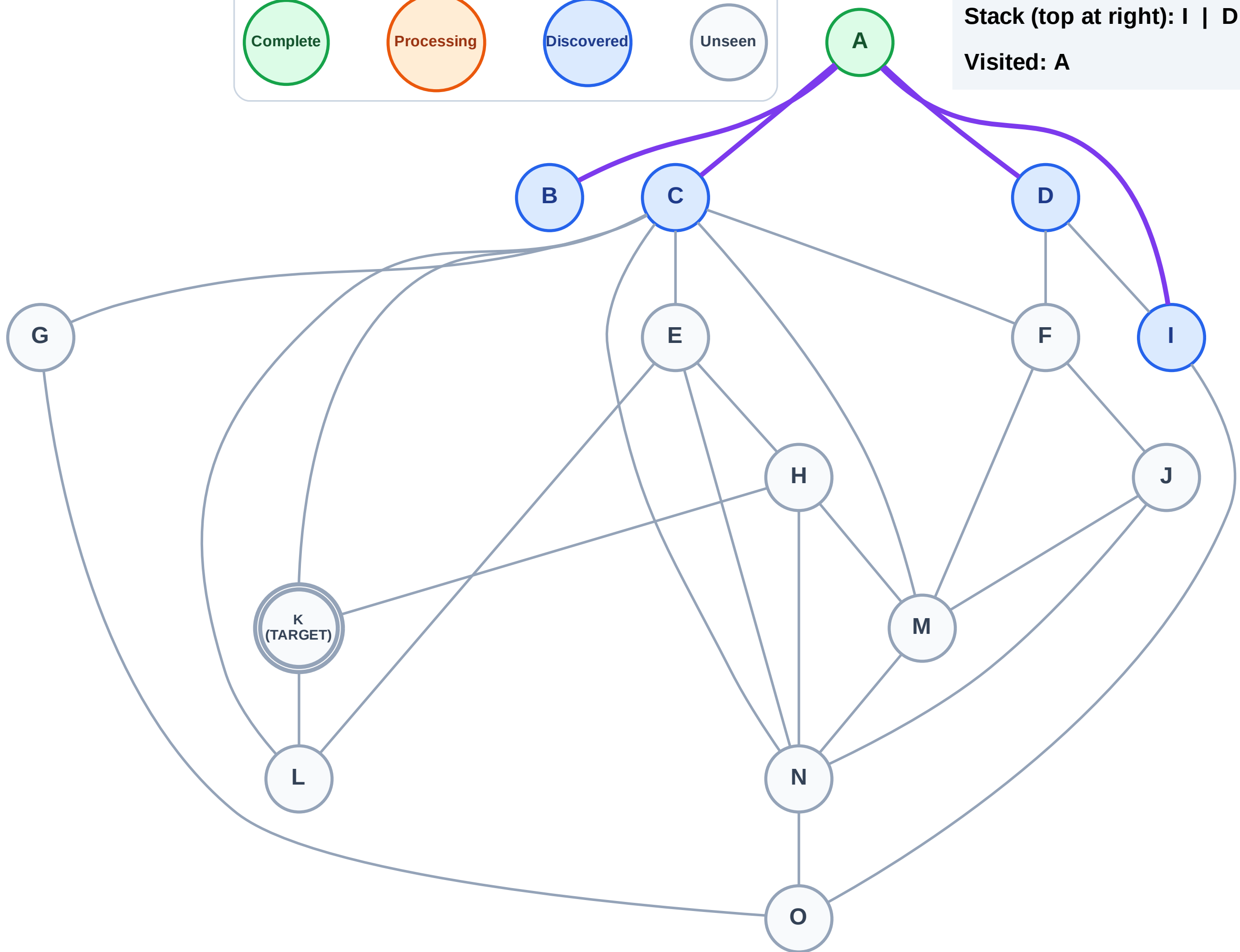
Step 3: Discover I, D, C, B from A

Legend



Stack (top at right): I | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

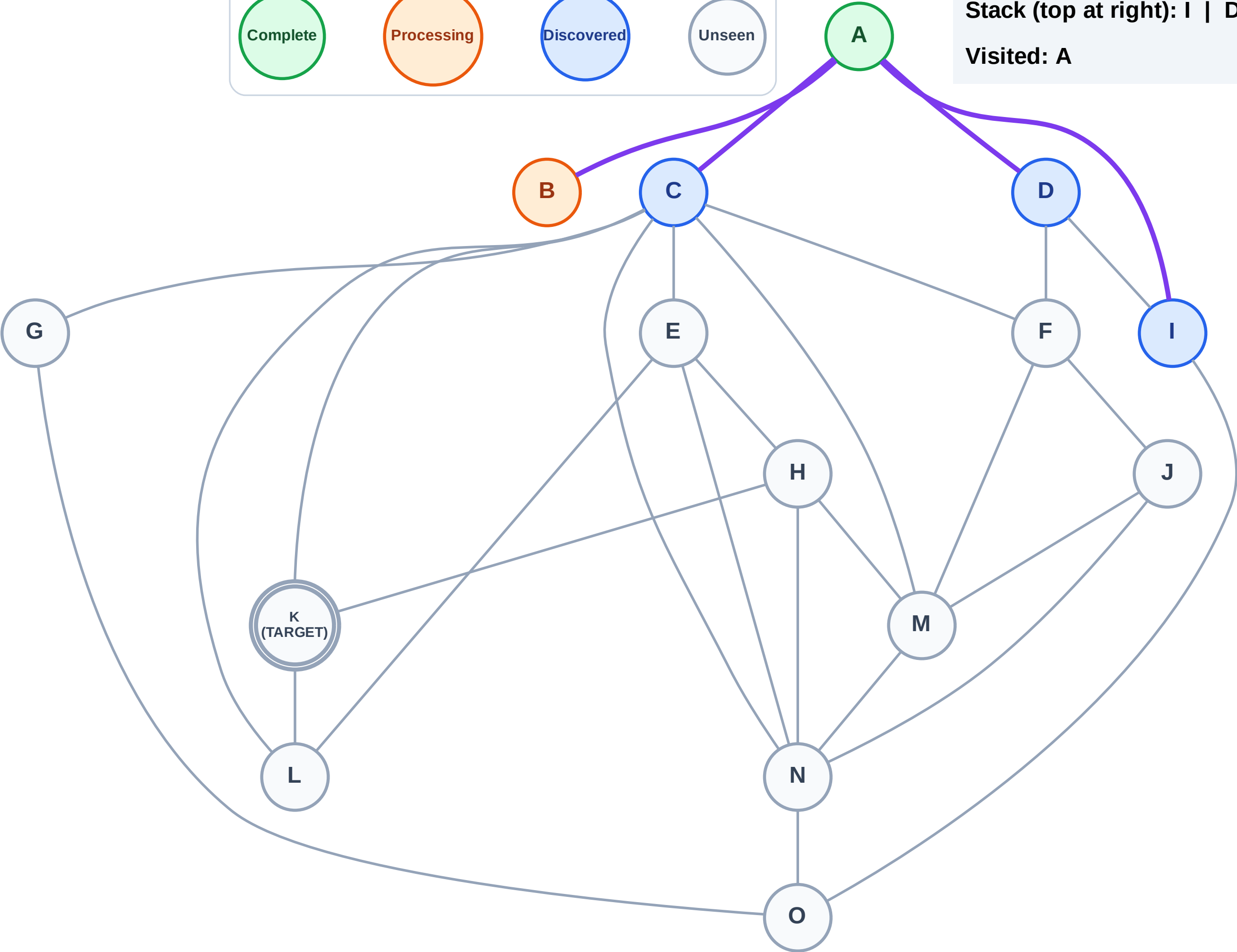
Processing

Discovered

Unseen

Stack (top at right): I | D | C

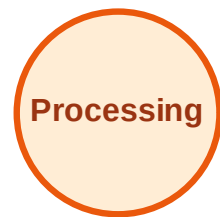
Visited: A



DFS Traversal

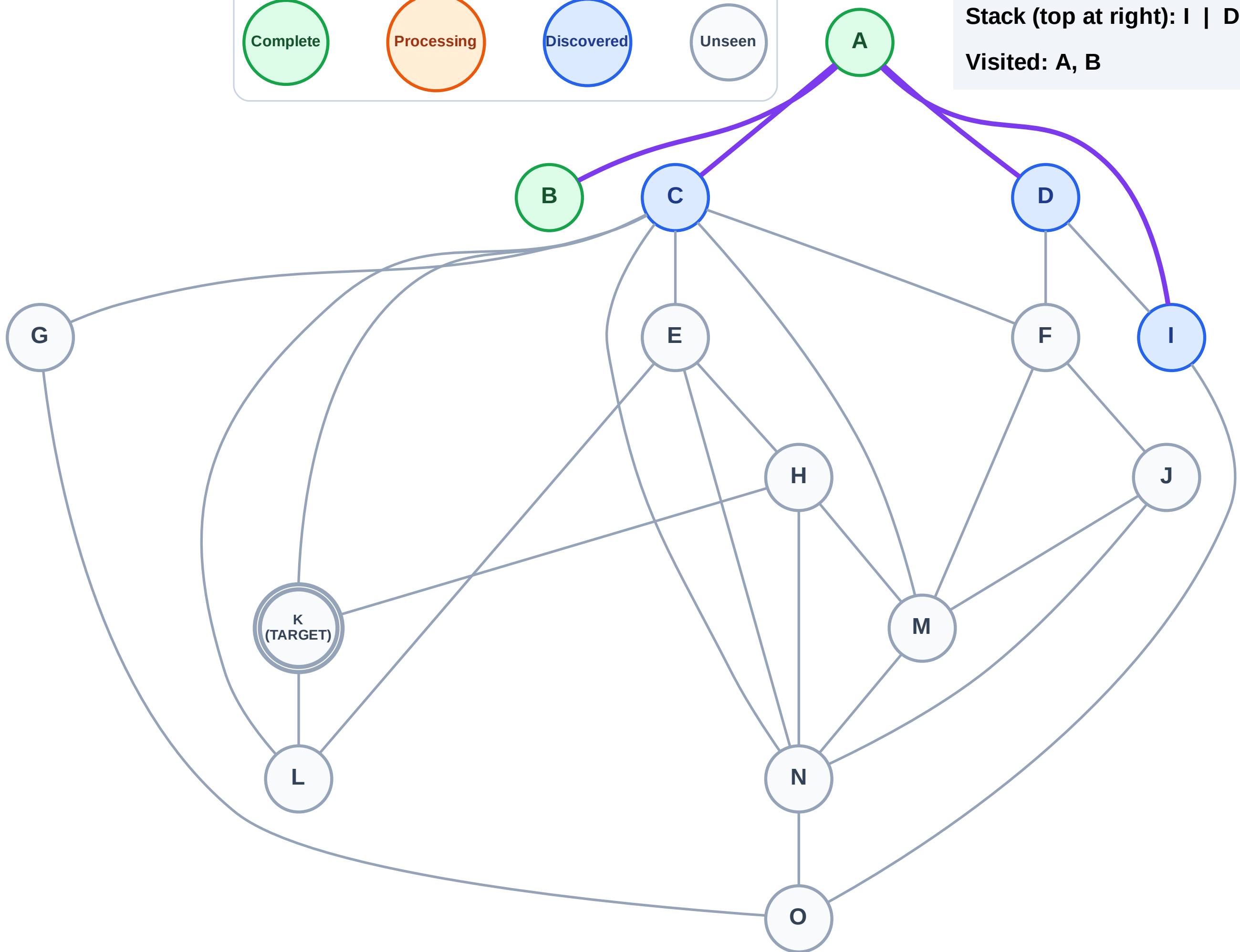
Step 5: No new neighbors from B

Legend



Stack (top at right): I | D | C

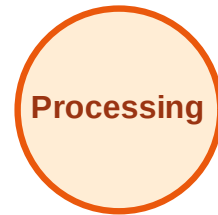
Visited: A, B



DFS Traversal

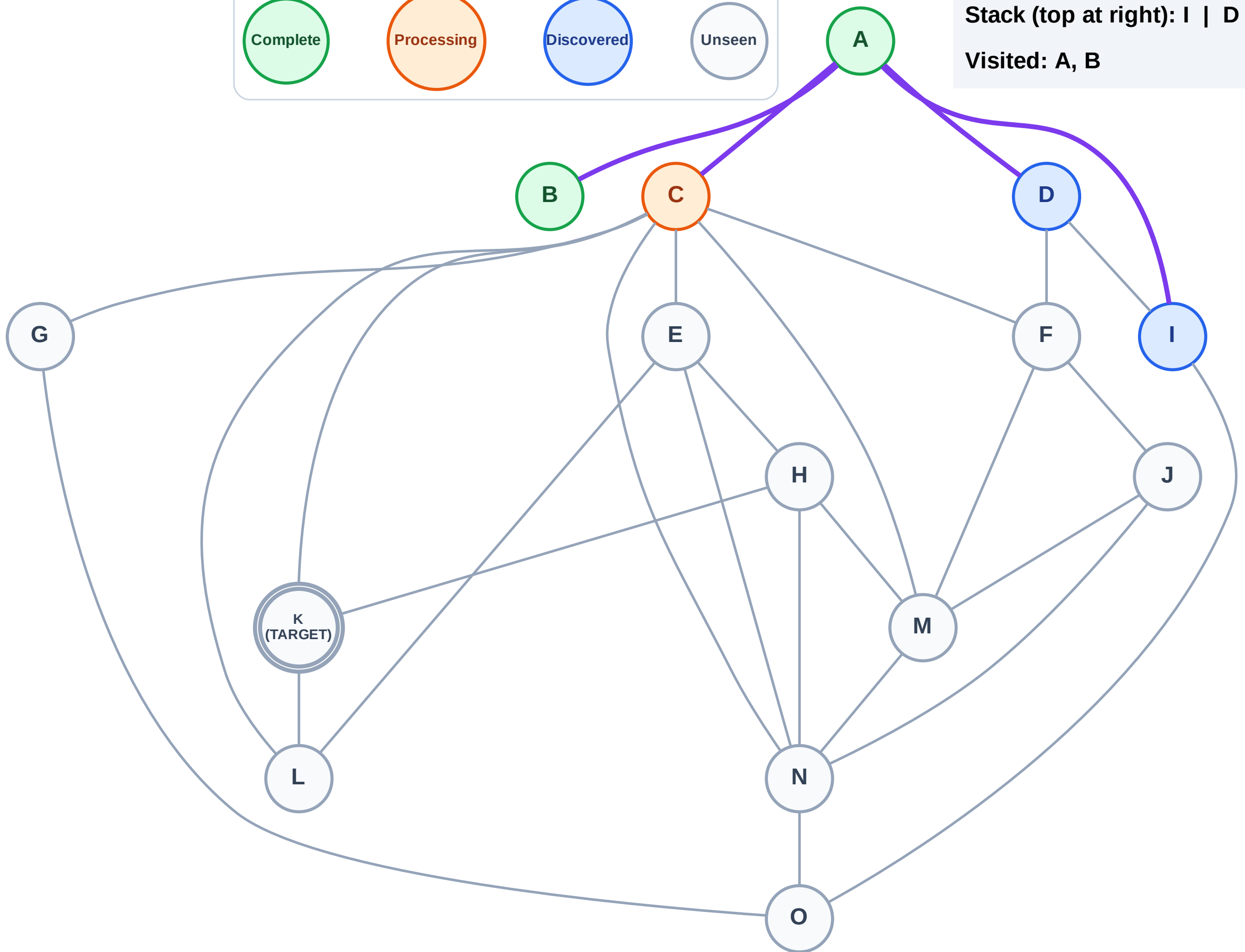
Step 6: Process node C

Legend



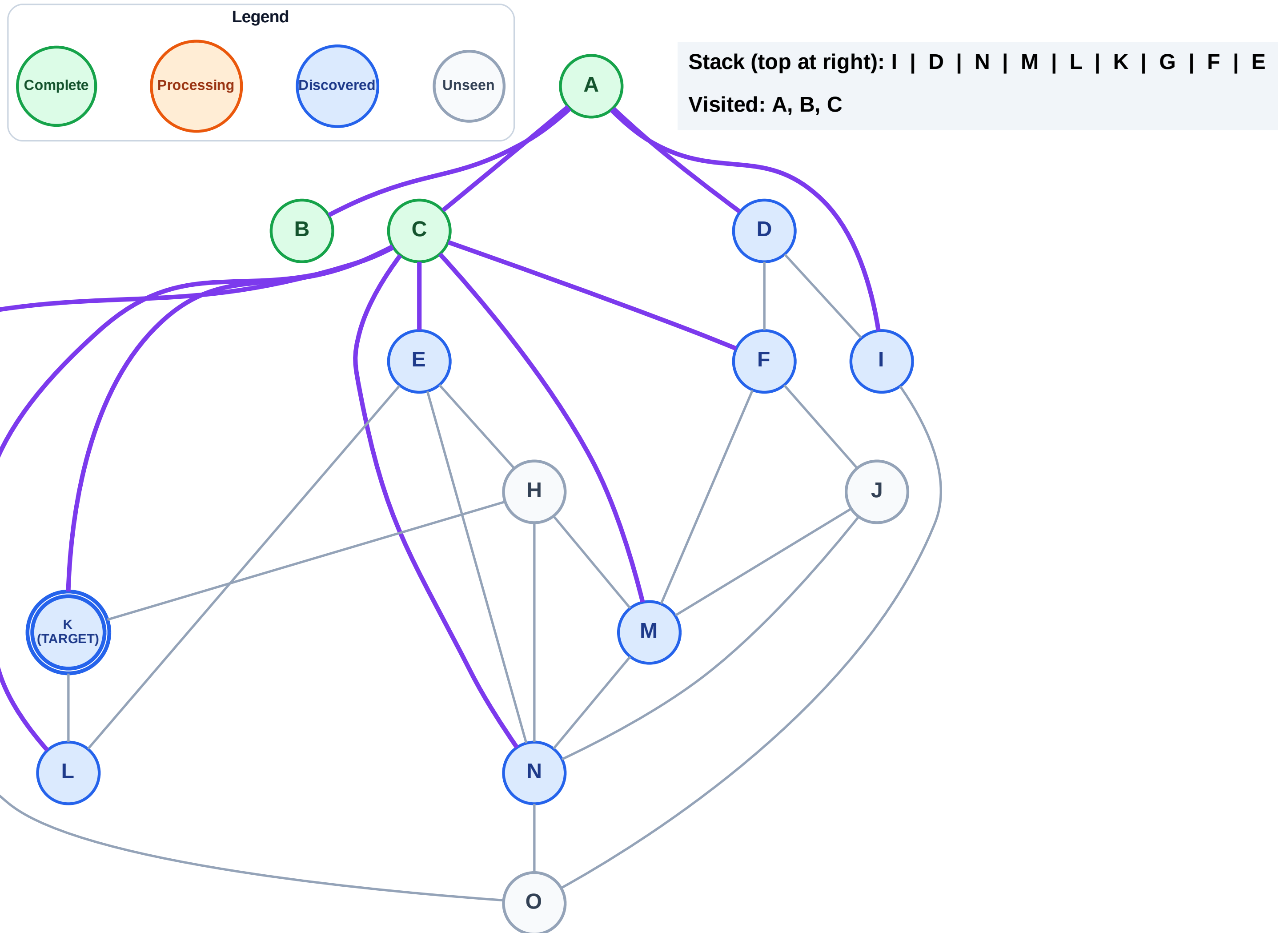
Stack (top at right): I | D

Visited: A, B



DFS Traversal

Step 7: Discover N, M, L, K, G, F, E from C



DFS Traversal

Step 8: Process node E

Legend

Complete

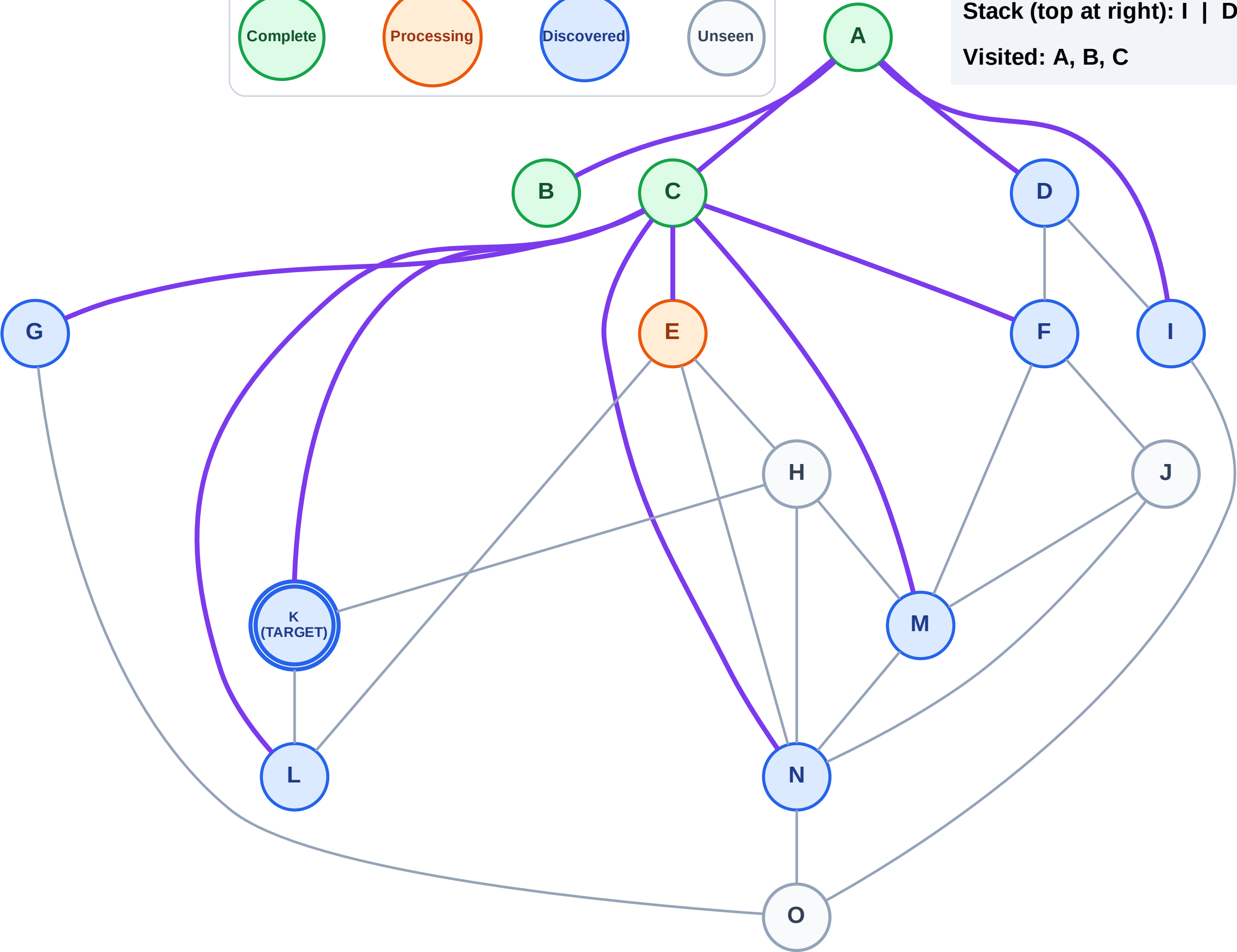
Processing

Discovered

Unseen

Stack (top at right): I | D | N | M | L | K | G | F

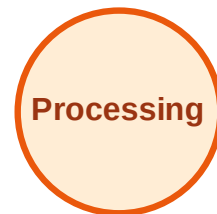
Visited: A, B, C



DFS Traversal

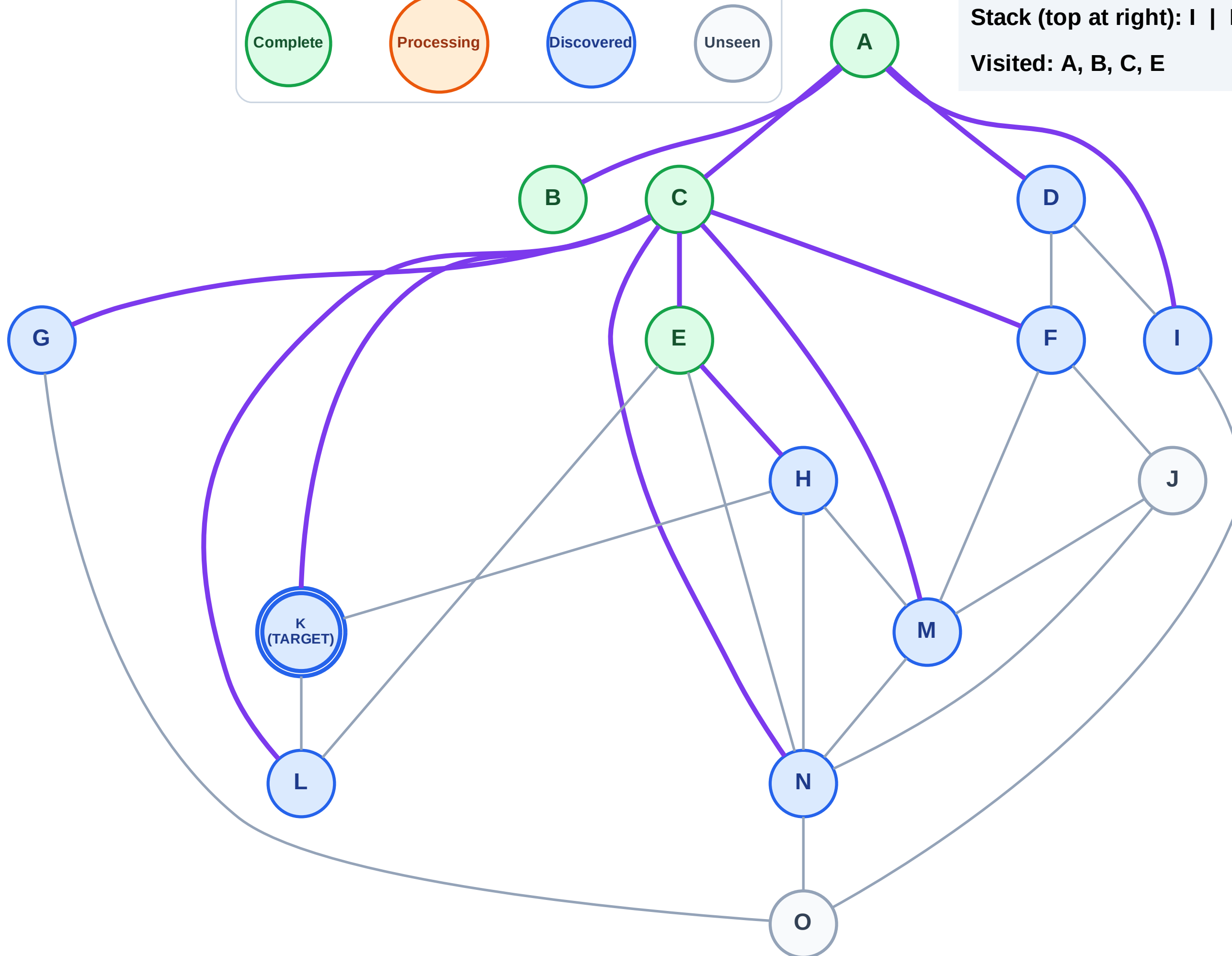
Step 9: Discover H from E

Legend



Stack (top at right): I | D | N | M | L | K | G | F | H

Visited: A, B, C, E



DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



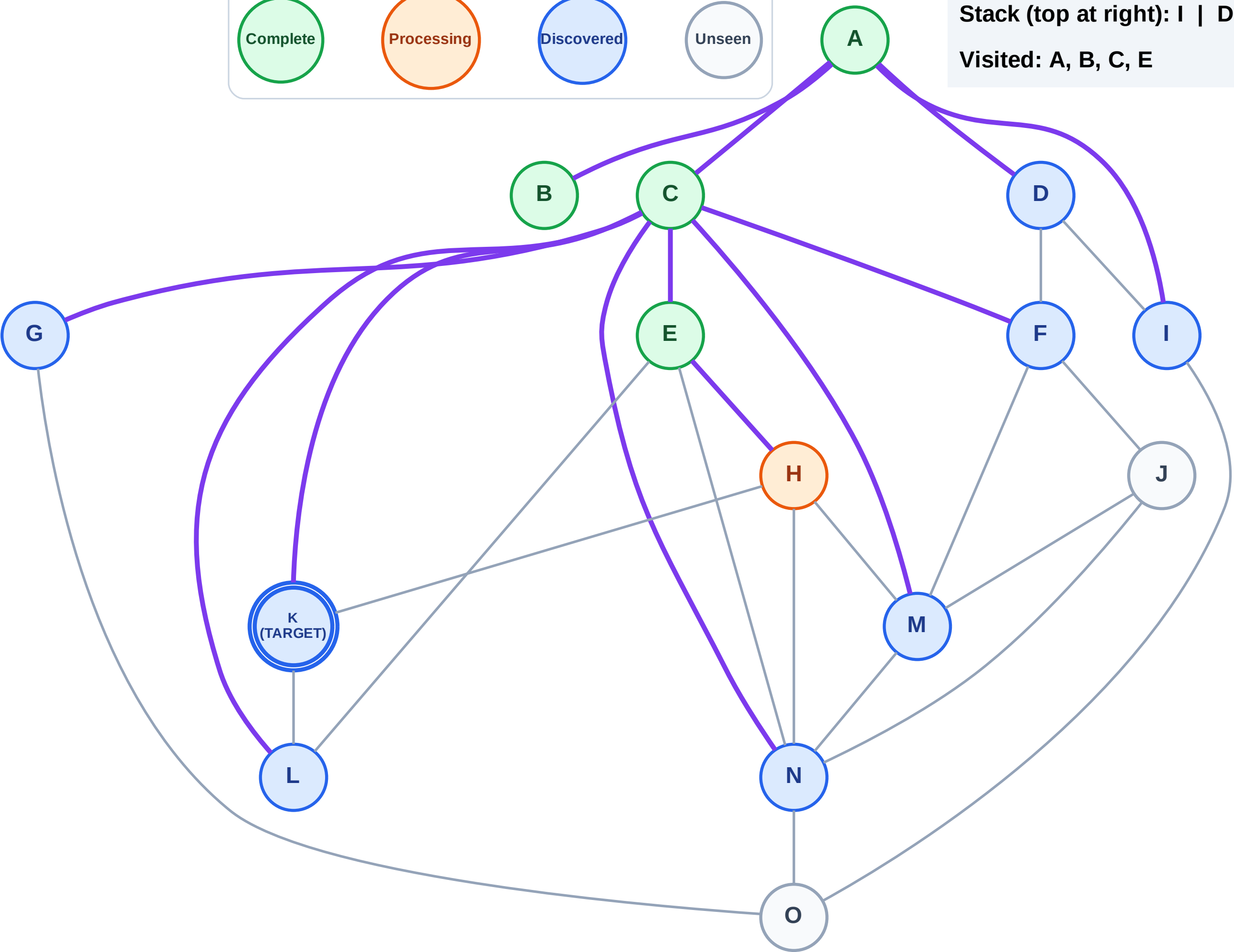
Discovered



Unseen

Stack (top at right): I | D | N | M | L | K | G | F

Visited: A, B, C, E



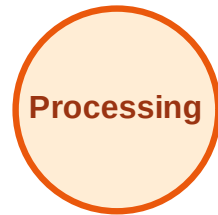
DFS Traversal

Step 11: No new neighbors from H

Legend



Complete



Processing



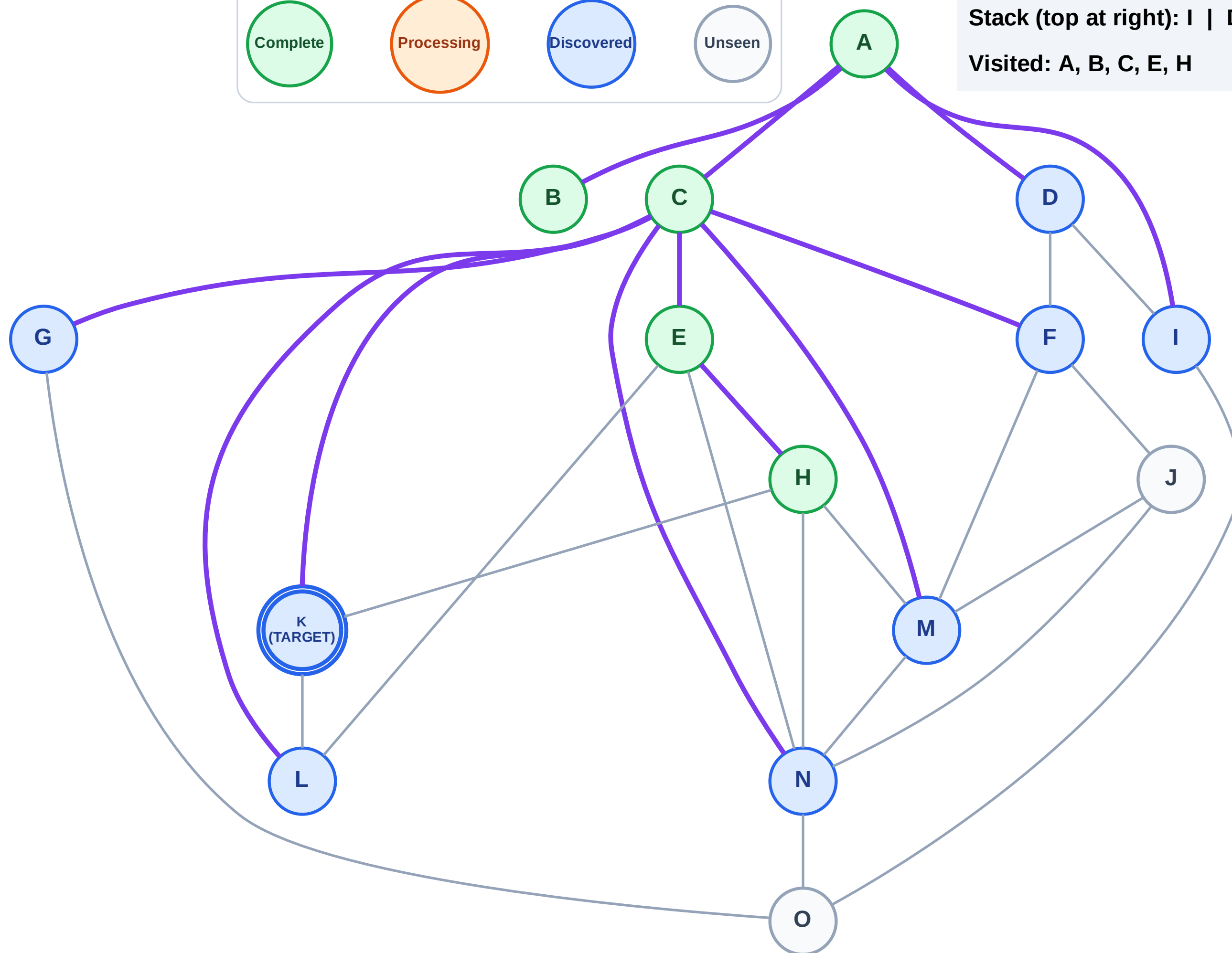
Discovered



Unseen

Stack (top at right): I | D | N | M | L | K | G | F

Visited: A, B, C, E, H



DFS Traversal

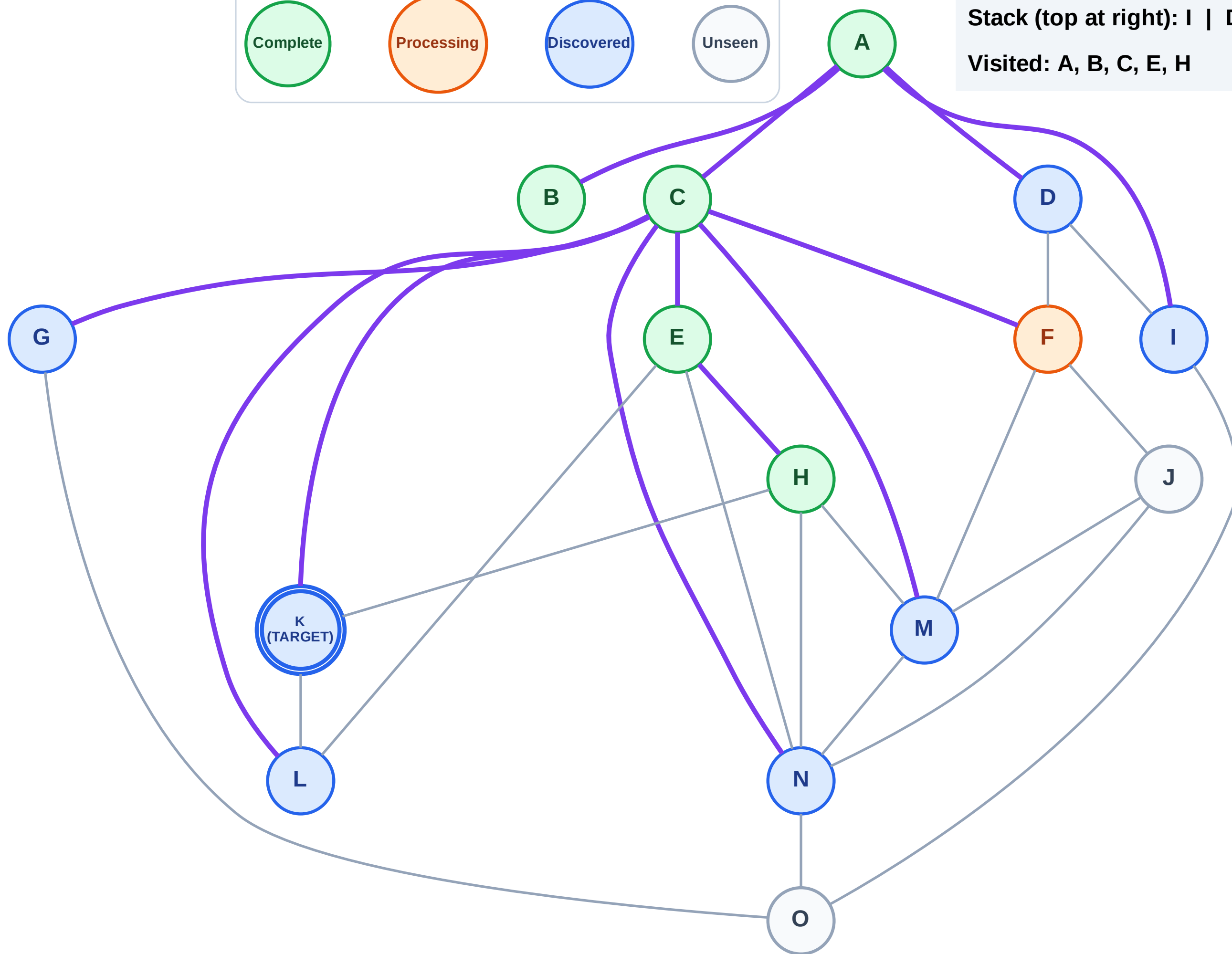
Step 12: Process node F

Legend



Stack (top at right): I | D | N | M | L | K | G

Visited: A, B, C, E, H



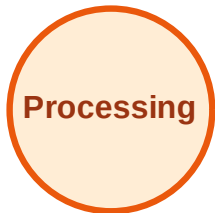
DFS Traversal

Step 13: Discover J from F

Legend



Complete



Processing



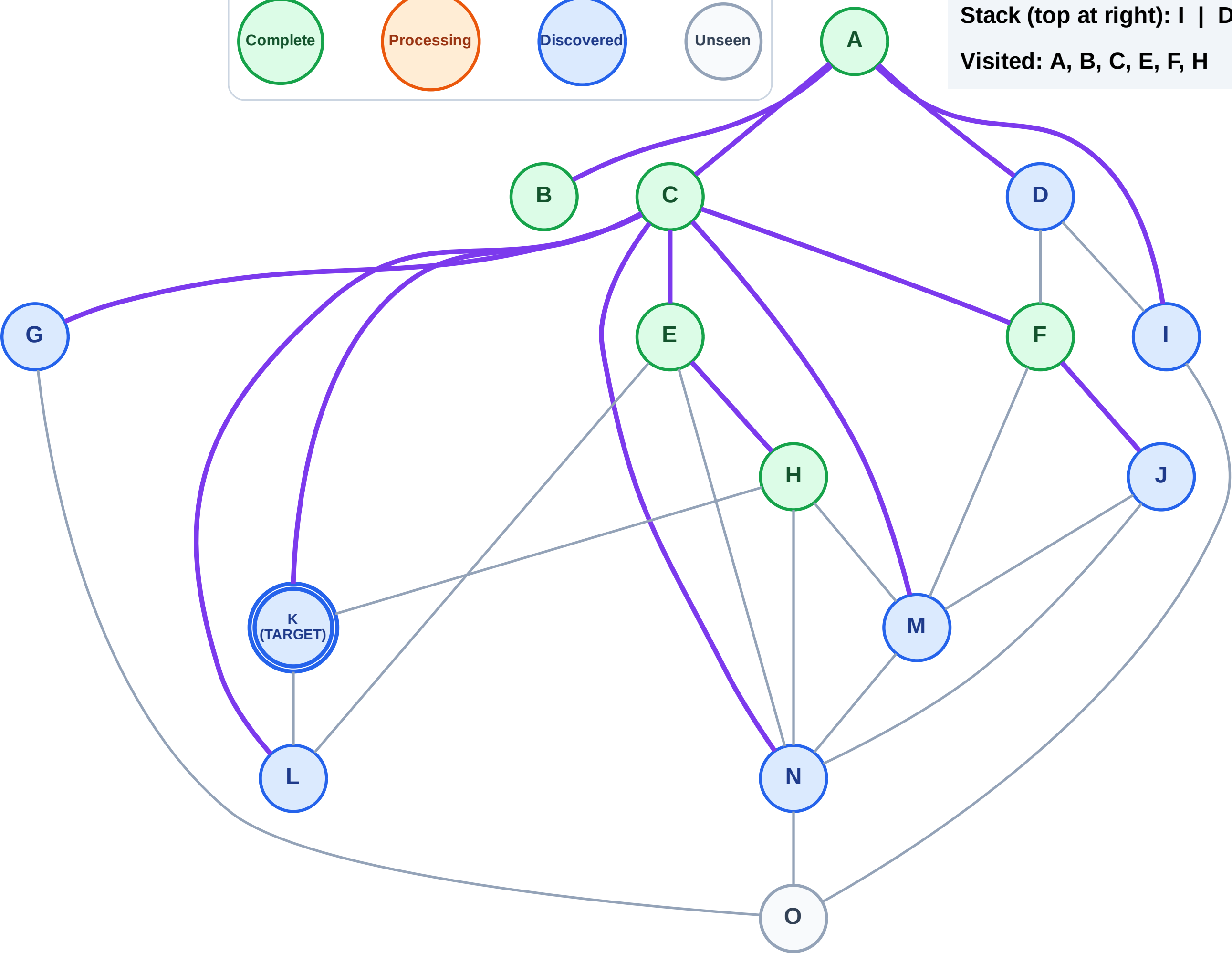
Discovered



Unseen

Stack (top at right): I | D | N | M | L | K | G | J

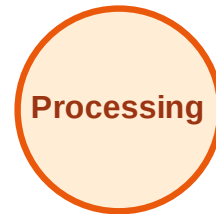
Visited: A, B, C, E, F, H



DFS Traversal

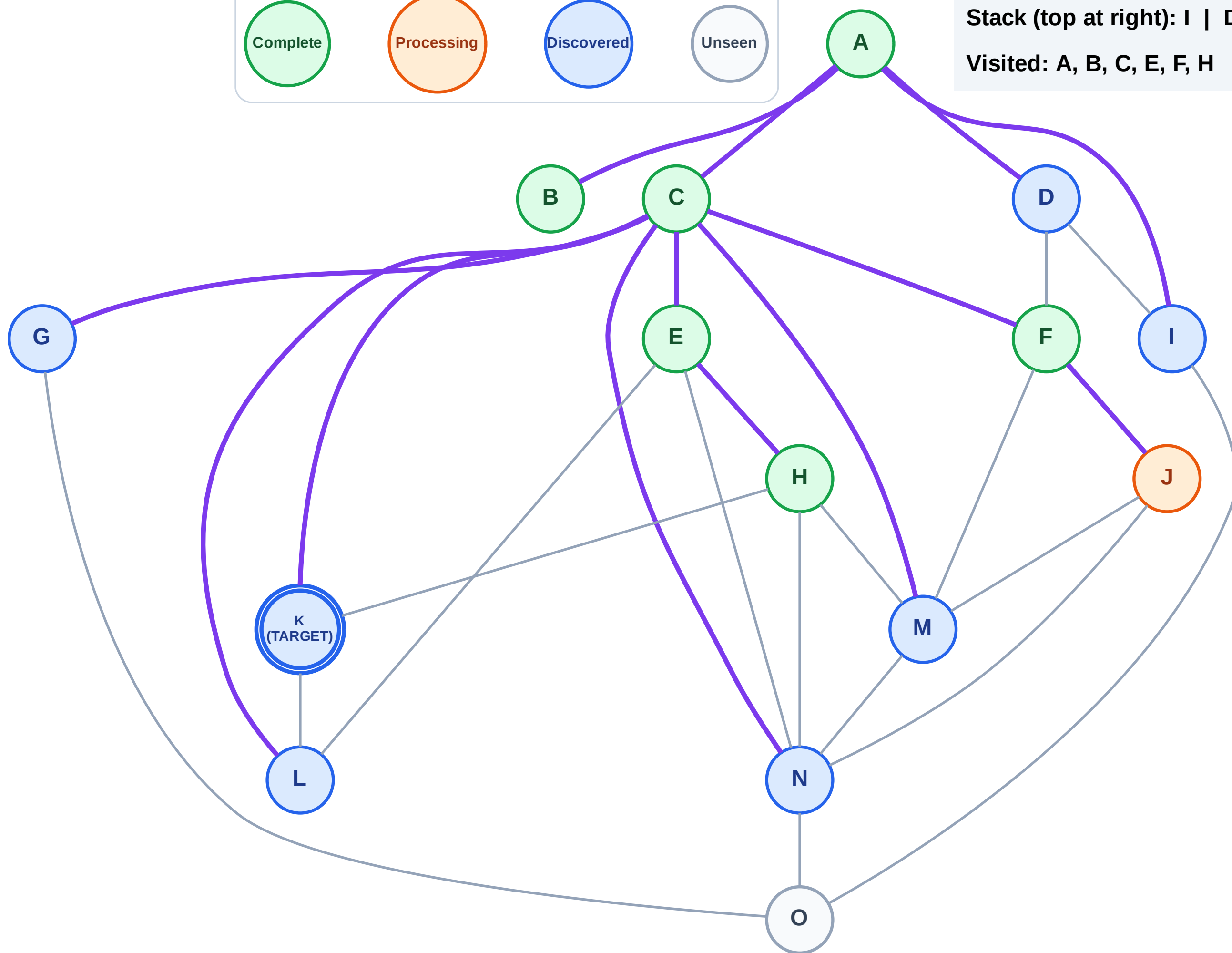
Step 14: Process node J

Legend



Stack (top at right): I | D | N | M | L | K | G

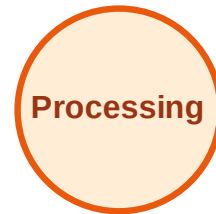
Visited: A, B, C, E, F, H



DFS Traversal

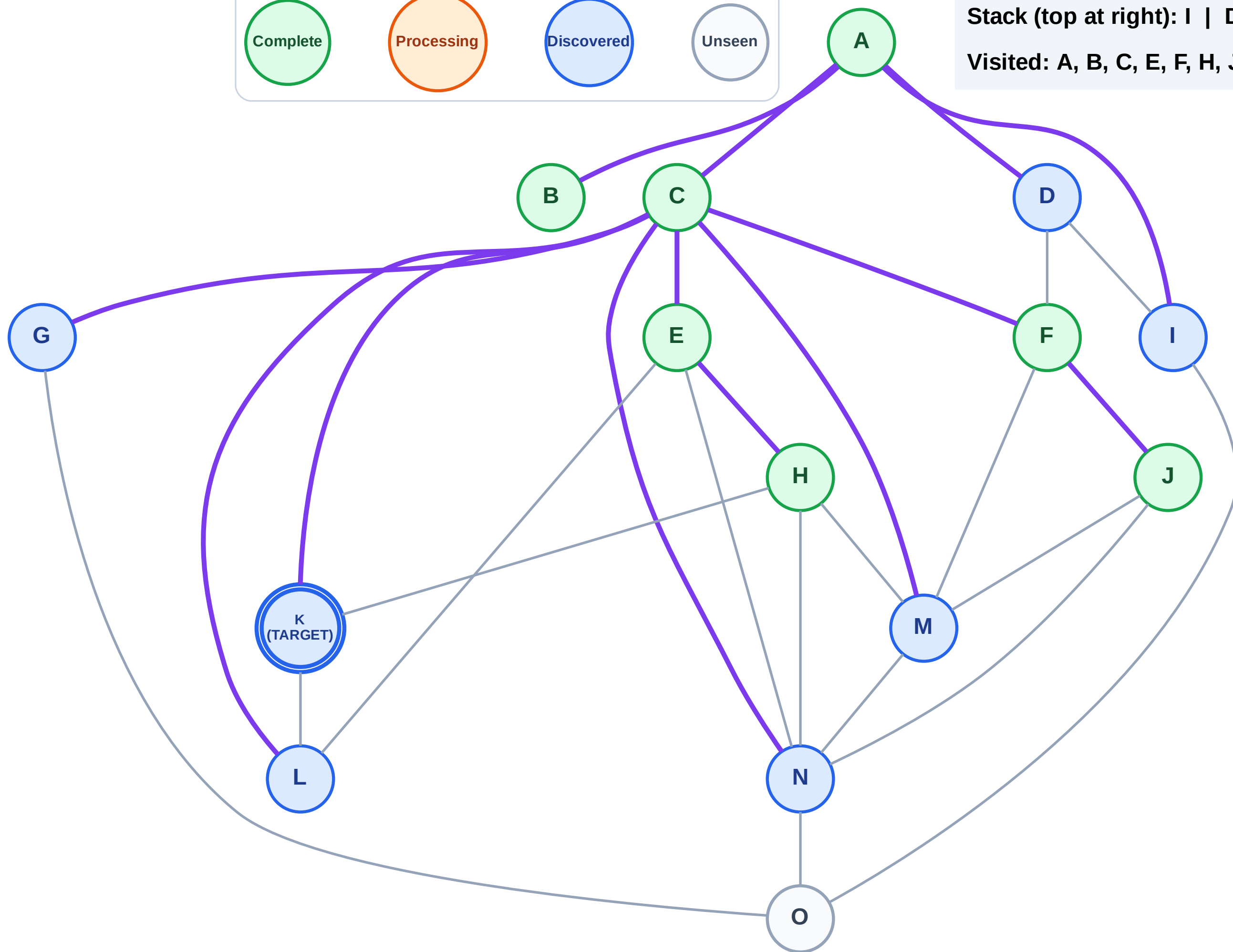
Step 15: No new neighbors from J

Legend



Stack (top at right): I | D | N | M | L | K | G

Visited: A, B, C, E, F, H, J



DFS Traversal

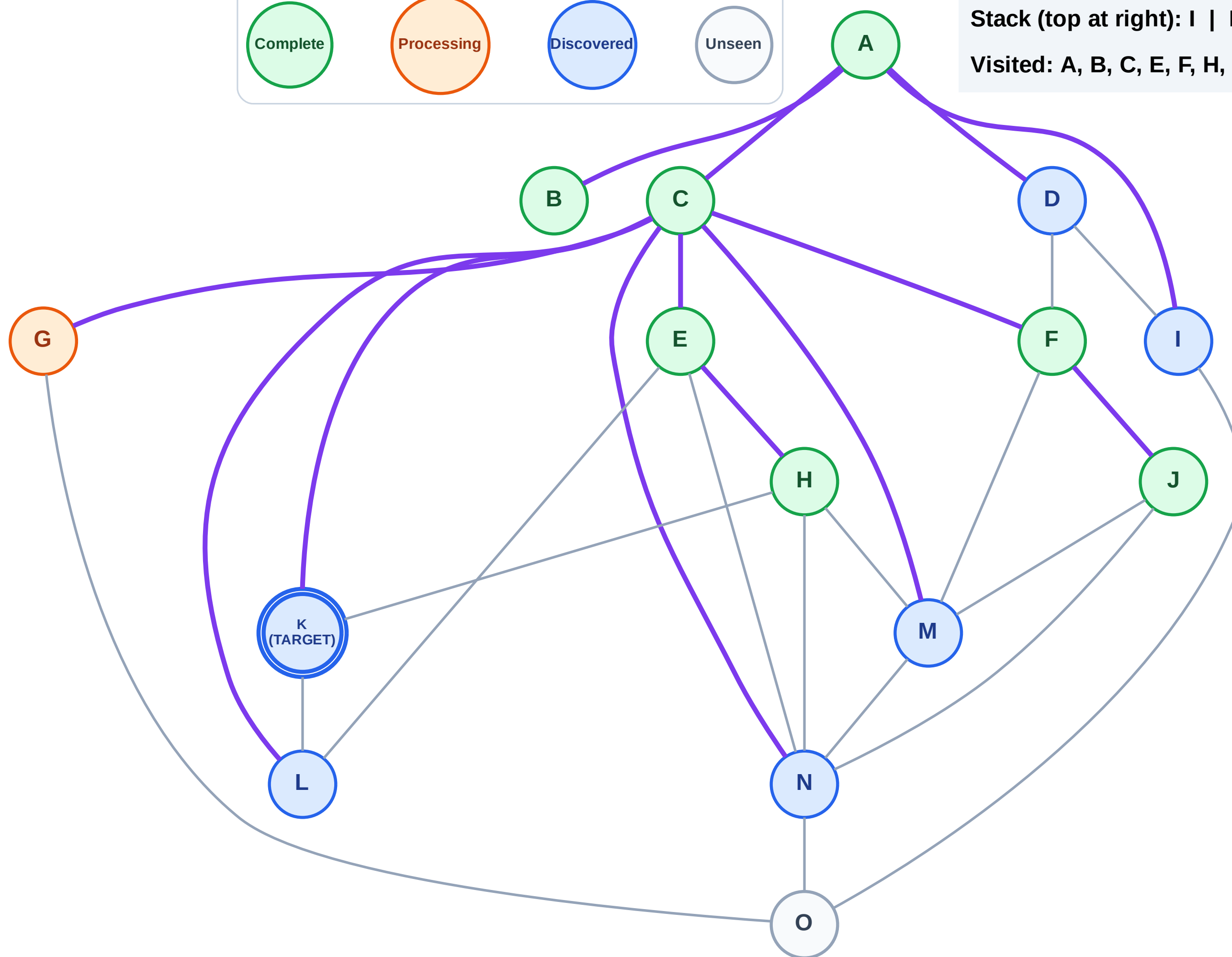
Step 16: Process node G

Legend



Stack (top at right): I | D | N | M | L | K

Visited: A, B, C, E, F, H, J



DFS Traversal

Step 17: Discover O from G

Legend

Complete

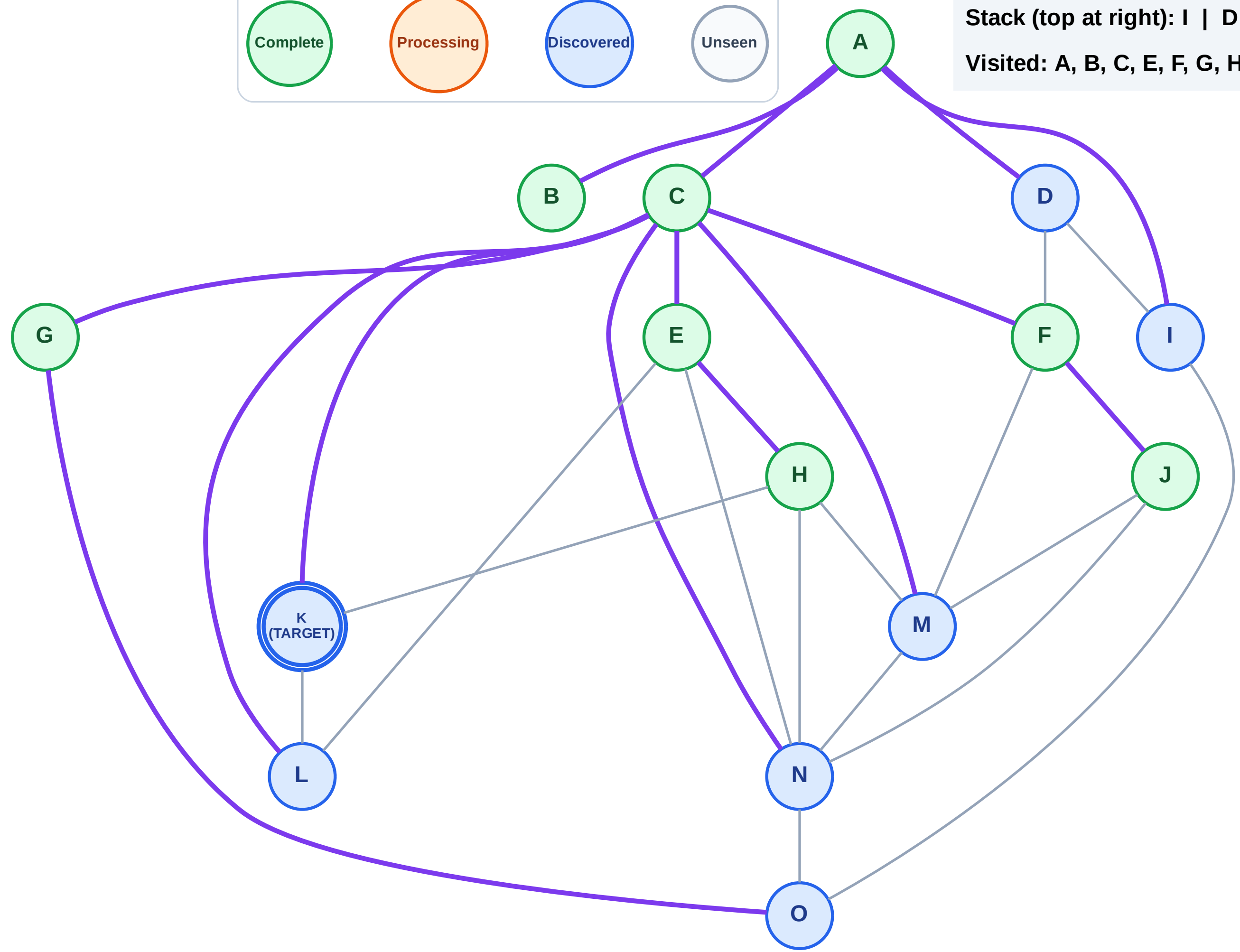
Processing

Discovered

Unseen

Stack (top at right): I | D | N | M | L | K | O

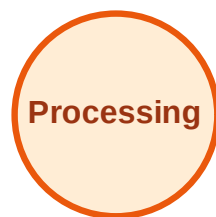
Visited: A, B, C, E, F, G, H, J



DFS Traversal

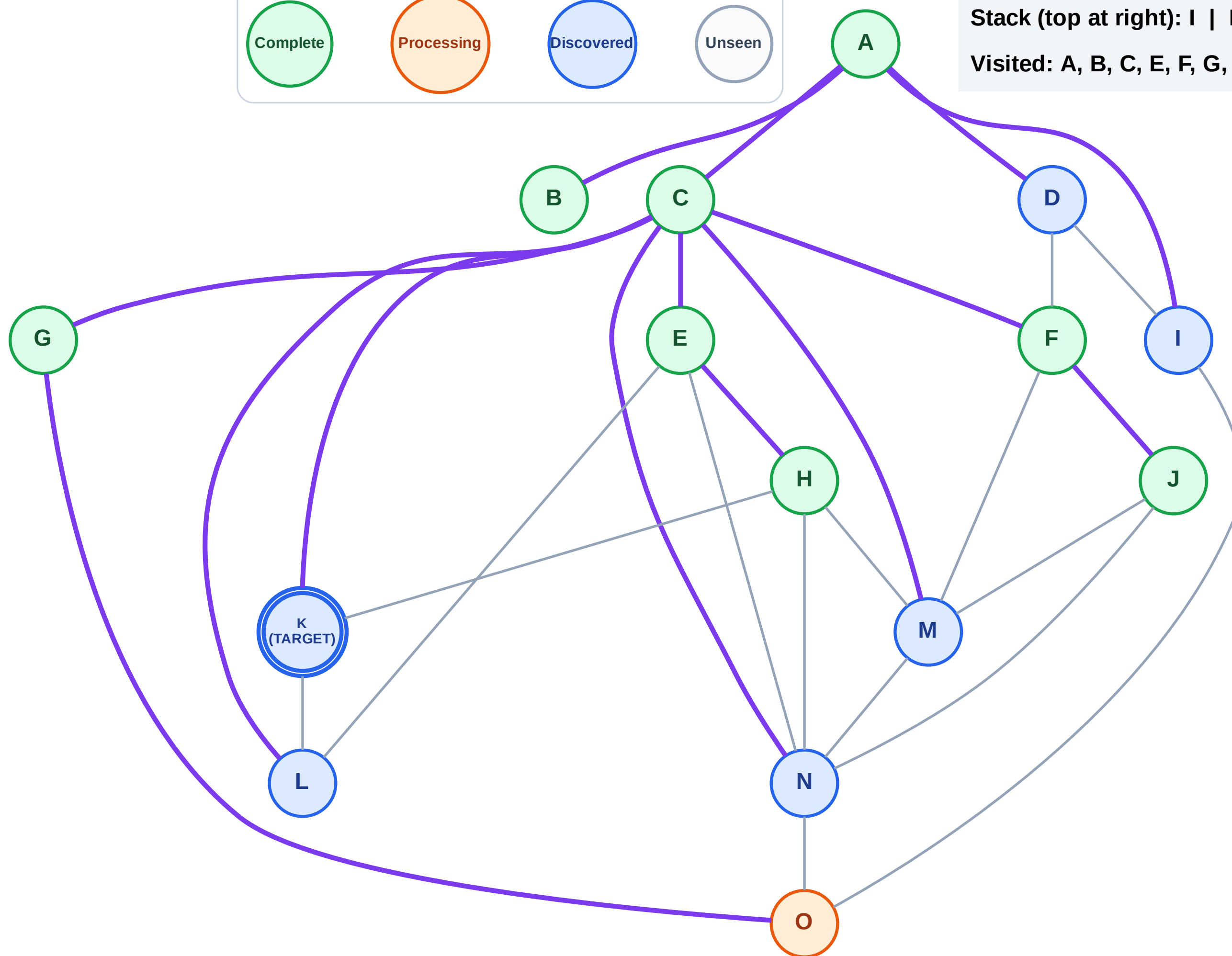
Step 18: Process node O

Legend



Stack (top at right): I | D | N | M | L | K

Visited: A, B, C, E, F, G, H, J



DFS Traversal

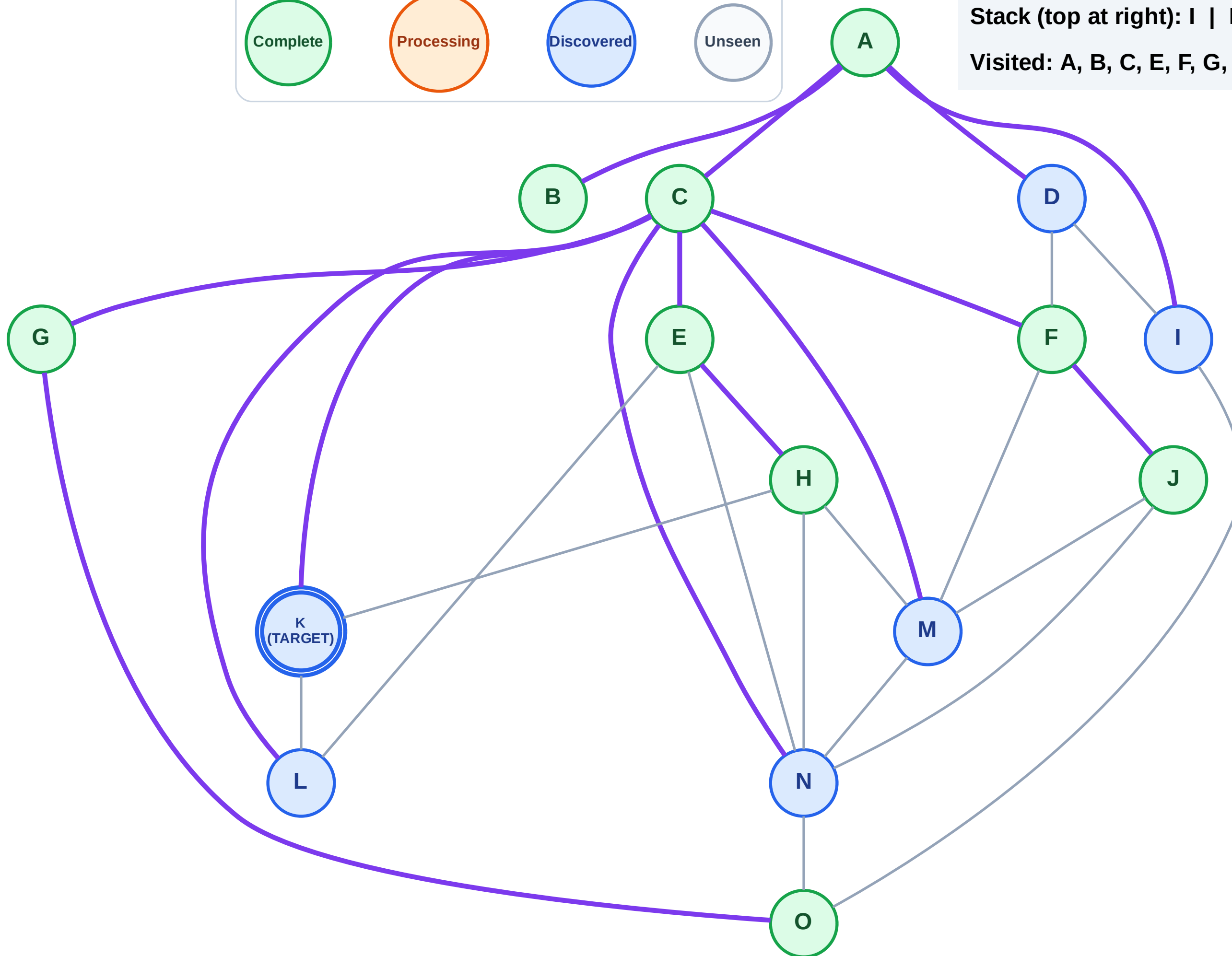
Step 19: No new neighbors from O

Legend



Stack (top at right): I | D | N | M | L | K

Visited: A, B, C, E, F, G, H, J, O



DFS Traversal

Step 20: Process node K

Legend

Complete

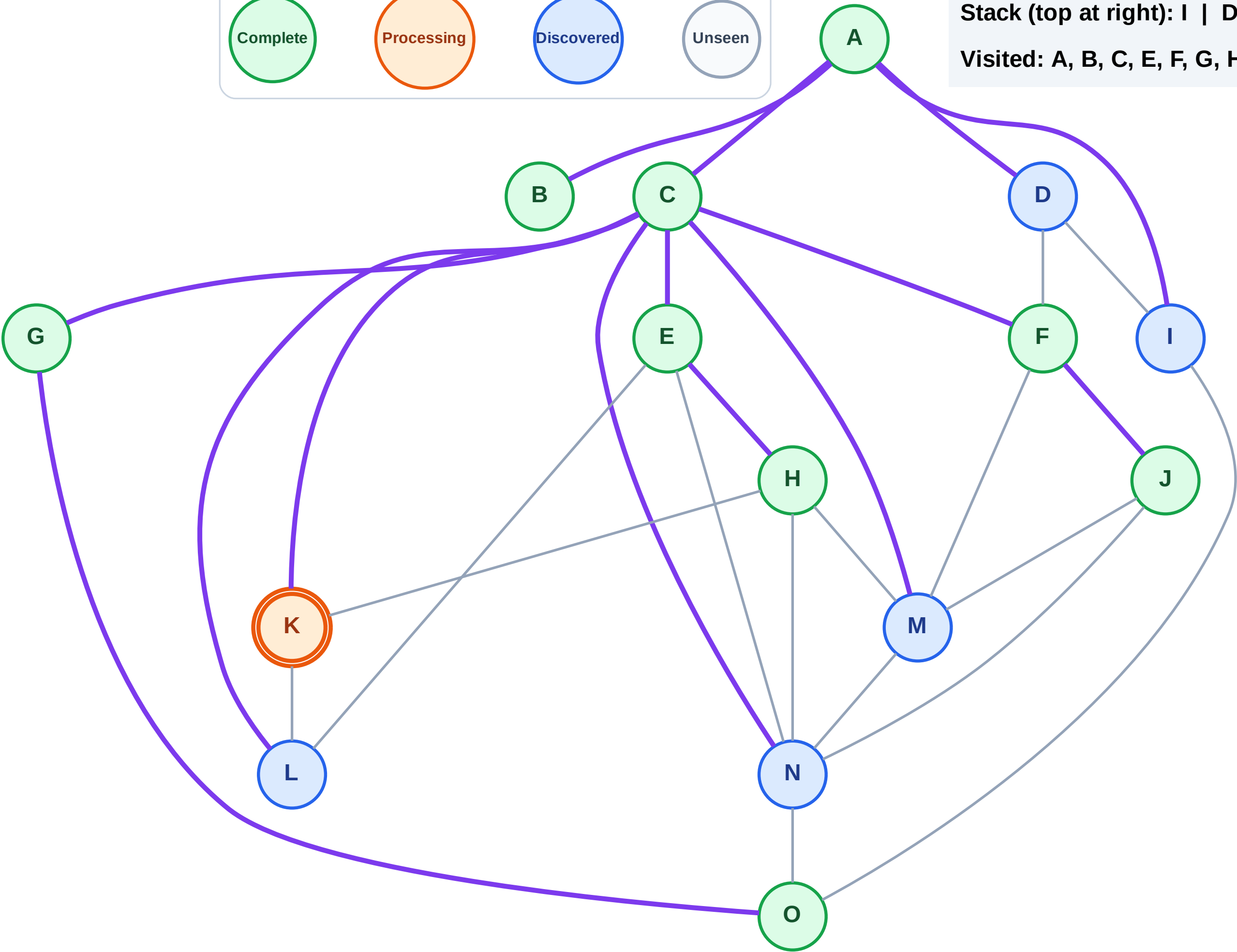
Processing

Discovered

Unseen

Stack (top at right): I | D | N | M | L

Visited: A, B, C, E, F, G, H, J, O



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): I | D | N | M | L
Visited: A, B, C, E, F, G, H, J, K, O

